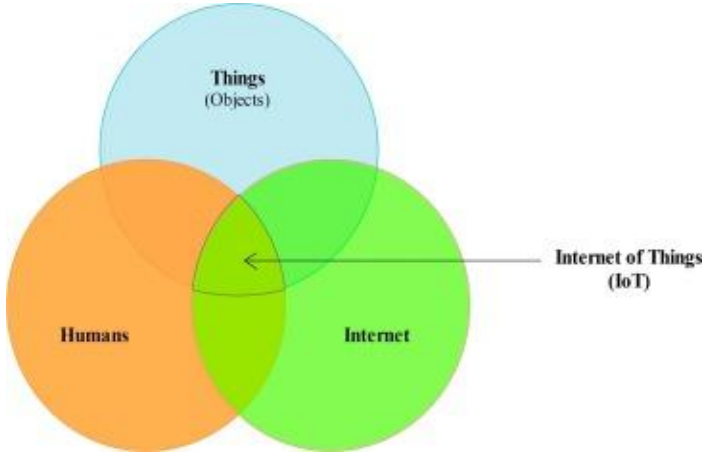
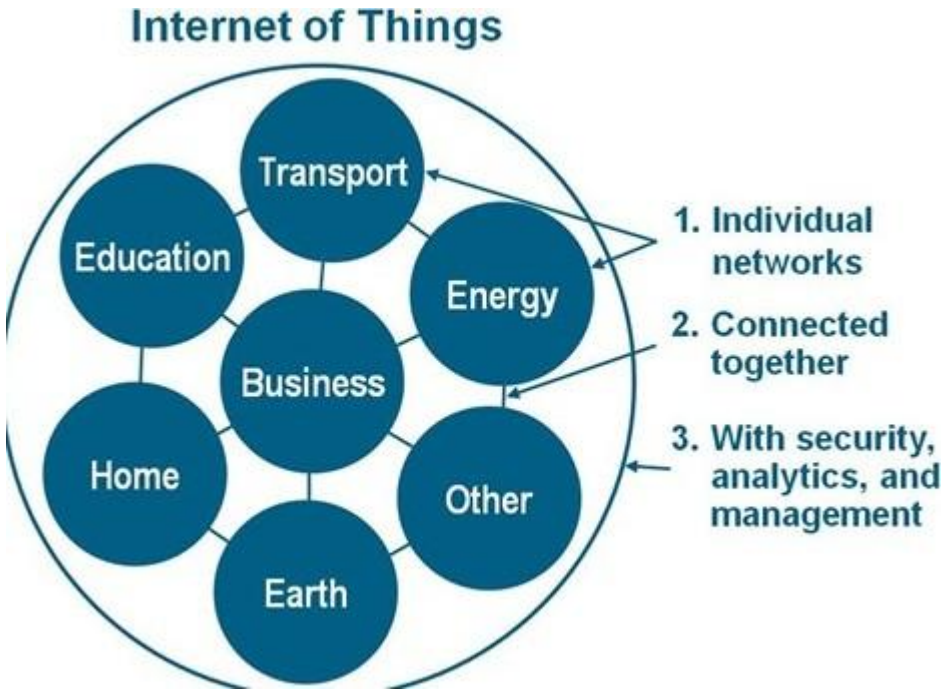
Introduction to IoT and Smart System



Source: https://doi.org/10.1016/j.compag.2018.12.039

Smart Device

- **Internet of Things (IoT)** is a massive network of **physical devices** embedded with sensors, software, electronics, and network which allows the devices to exchange or collect data and perform certain actions.



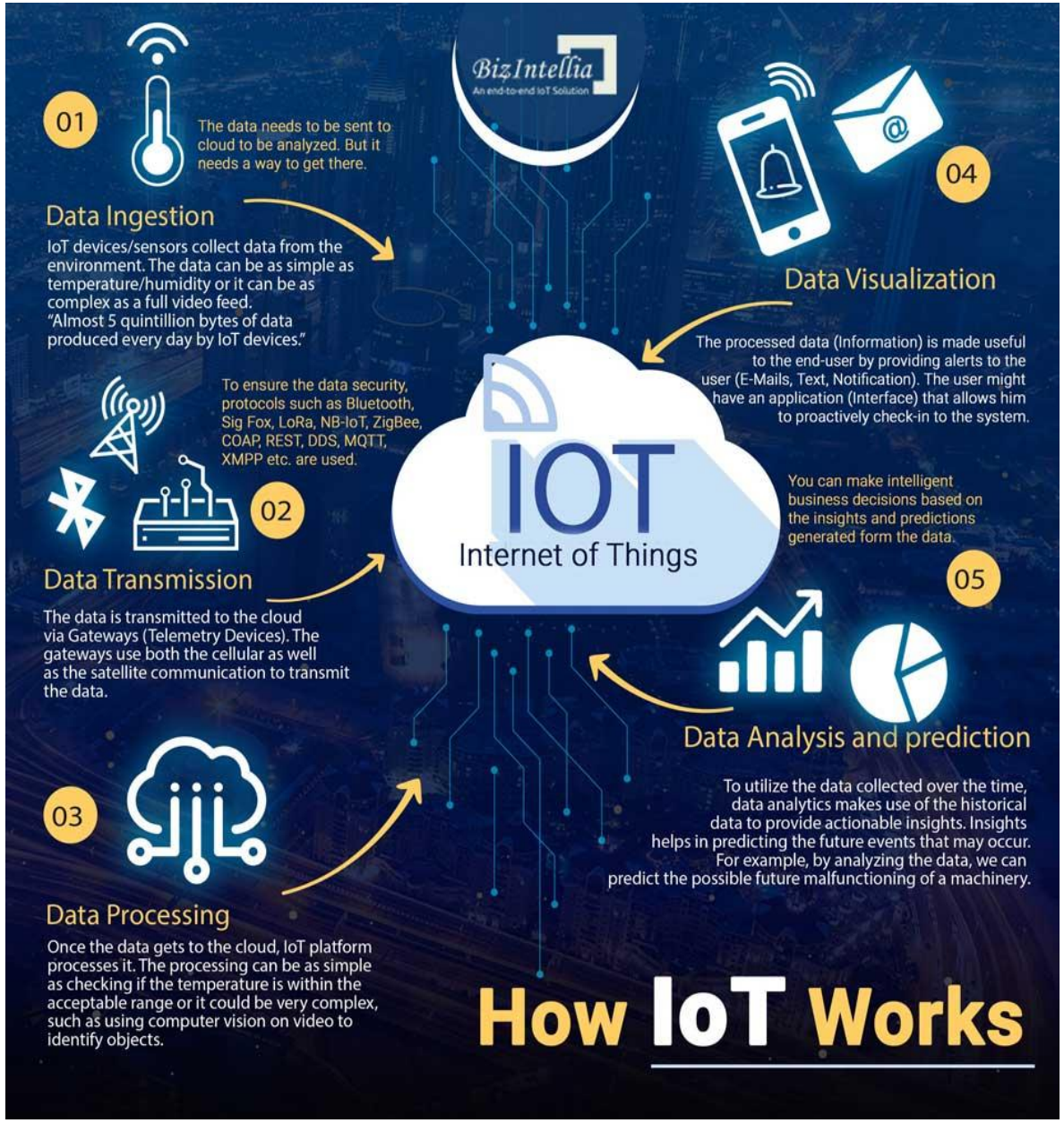
Source: Cisco IBSG, April 2011

Source: https://iotdunia.com/importance-of-iot-why-do-we-need-internet-of-things/rbsnp/

Have you ever been in a situation where you went to the market and forgot to switch off your AC, fan, or light and felt helpless that you cannot return home to switch it off?

This is where **IoT comes into the picture**

- It can remind you of the essential tasks that you often forget to do.
- Also, using the mobile application, you can access your home appliances integrated with IoT from anywhere around the world



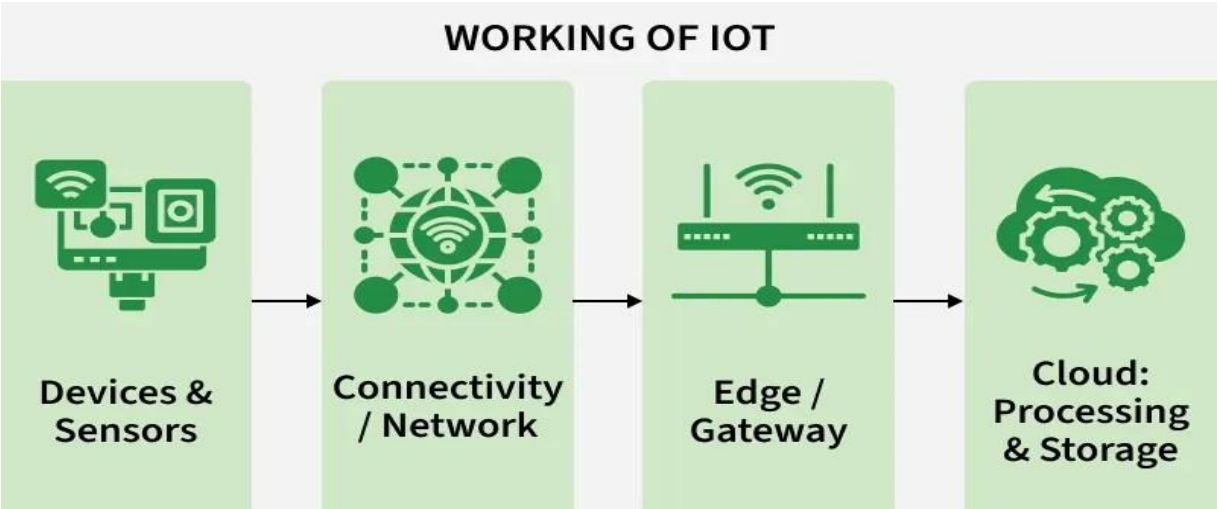
Source: https://www.biz4intellia.com/blog/simplified-insight-into-how-iot-works/

Introduction of internet of things (IOT)

Physical devices, appliances, gadgets, etc.

Internet of things

Through which these devices are connected



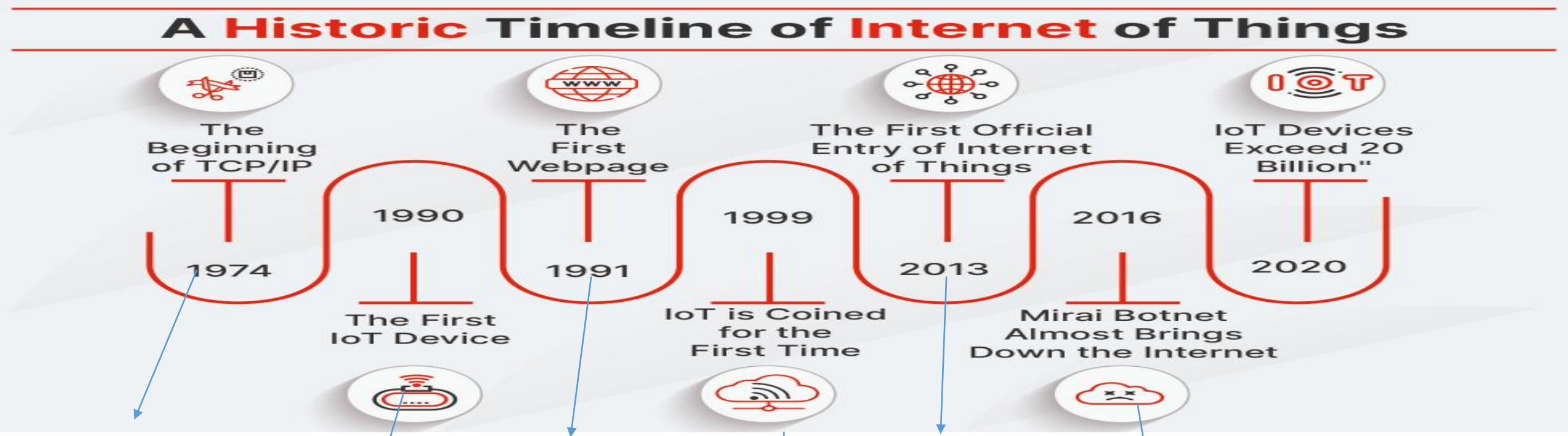
Source: https://www.geeksforgeeks.org/computer-networks/introduction-to-internet-of-things-iot-set-1/

Example: Smart Watch(sensor) → Mobile Phone (Gateway) → Cloud Server → User App

- In recent years, the Internet of Things (IoT) has experienced notable progress, fundamentally transforming the interactions with technology with our environment.
- IoT has rapidly changed, making it easier to make decisions and giving people new ways to use technology.
- Integrating intelligent gadgets and automation technology has transformed all facets of our existence, encompassing healthcare services, manufacturing, agriculture, and mining.

Source: Kramp, Thorsten, Rob Van Kranenburg, and Sebastian Lange. "Introduction to the Internet of Things." In *Enabling things to talk: Designing IoT solutions with the IoT architectural reference model*, pp. 1-10. Berlin, Heidelberg: Springer Berlin Heidelberg, 2013.

Progress of Internet of Things



Robert E. Kahn and Vinton Cerf

IP makes sure data goes from one place to another without any being lost along the way

TCP helps reassemble the data into what it was before it began its journey

John Romkey made a toaster that could be switched on and off over the internet

Tim Berners-Lee created the first webpage

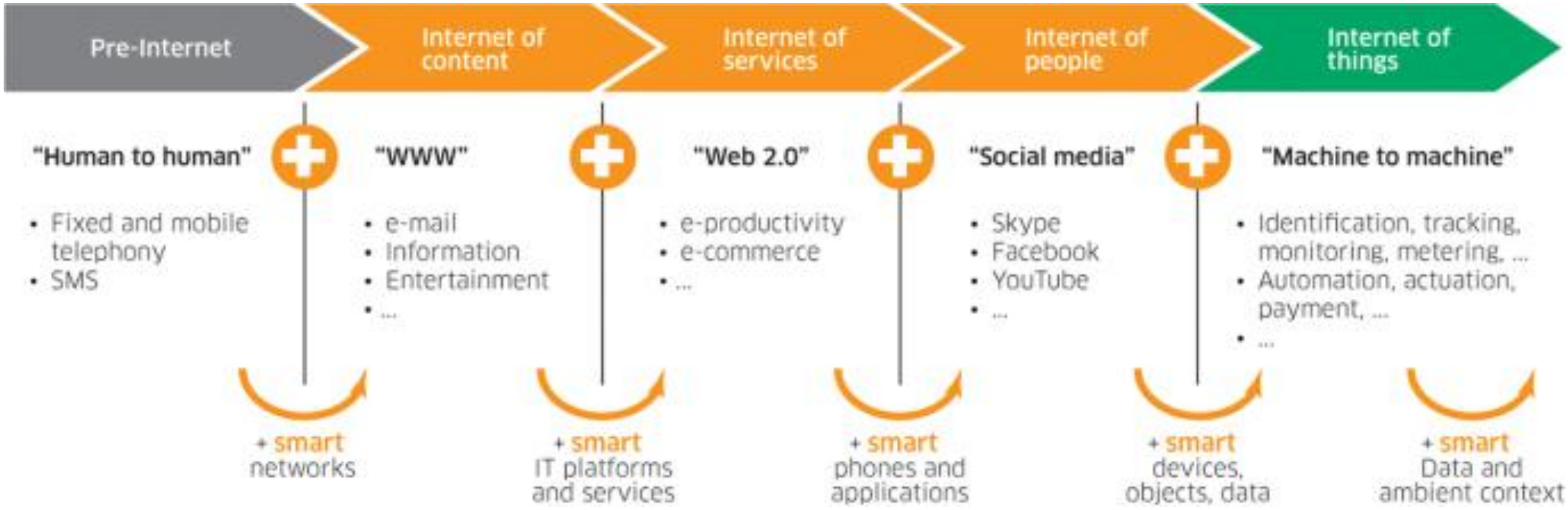
Kevin Ashton first coined the actual term "Internet of Things" in 1999 while he was giving a PowerPoint presentation for Procter & Gamble. He proposed devices that could sense their surroundings using sensors and software.

Even though the term had already been around for 14 years, the Oxford Dictionary added "Internet of Things" to its pages in 2013.

The **Mirai Botnet** infected over 600,000 devices with **malware** in 2016 and then executed an enormous **distributed denial-of-service (DDoS) attack** using the devices

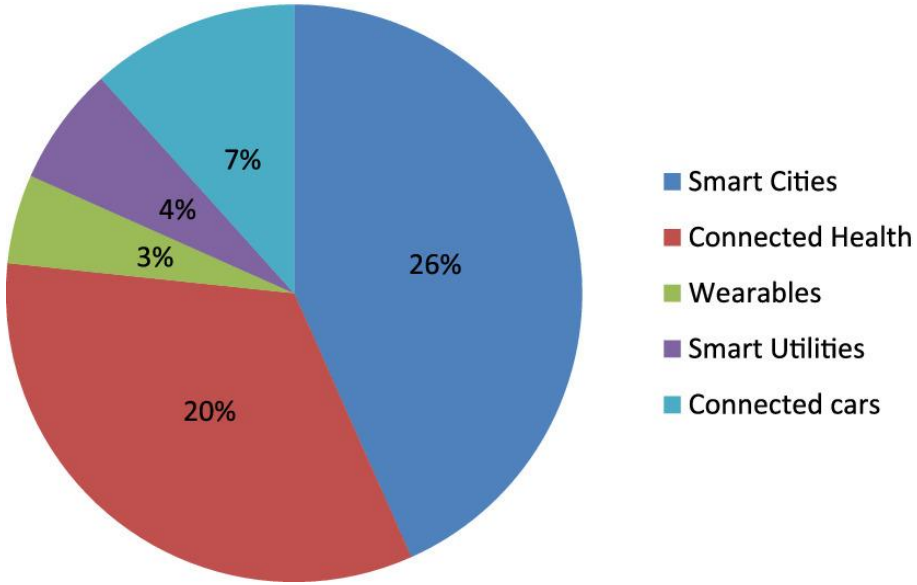
Source: https://www.fortinet.com/resources/cyberglossary/iot

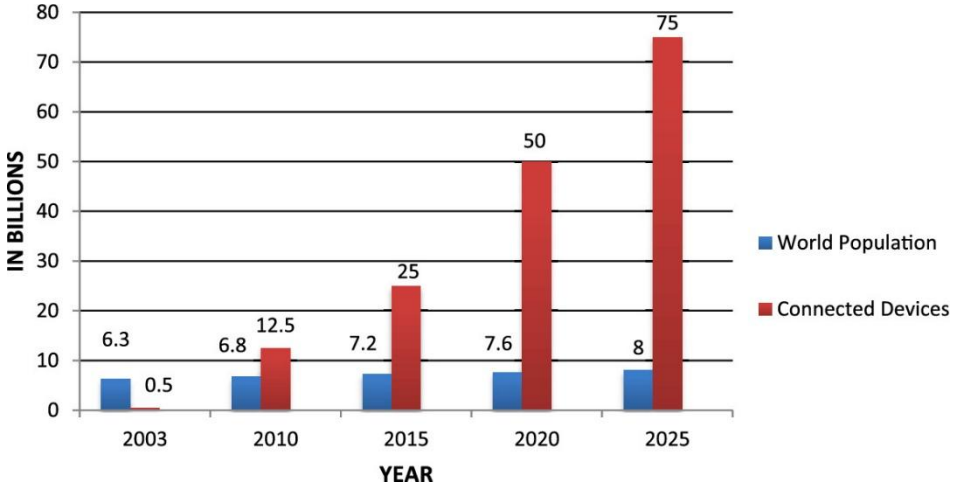
Development of Internet of Things



Source: https://doi.org/10.1016/j.compag.2018.12.039

Development of Internet of Things

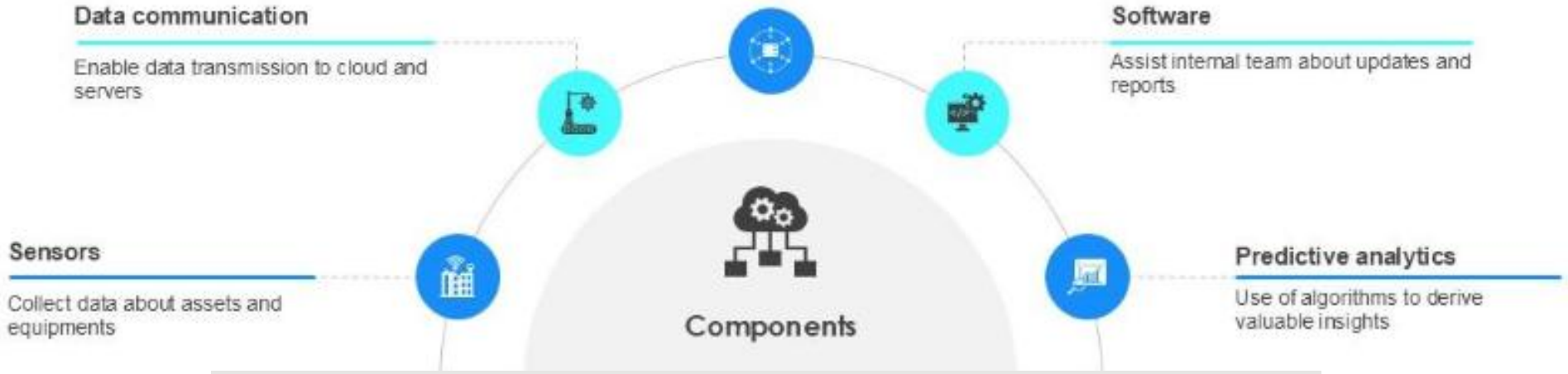




Source: https://onlinelibrary.wiley.com/doi/epdf/10.1002/ett.4166
Source: https://doi.org/10.1016/j.compag.2018.12.039
Source: https://doi.org/10.1002/ett.4166

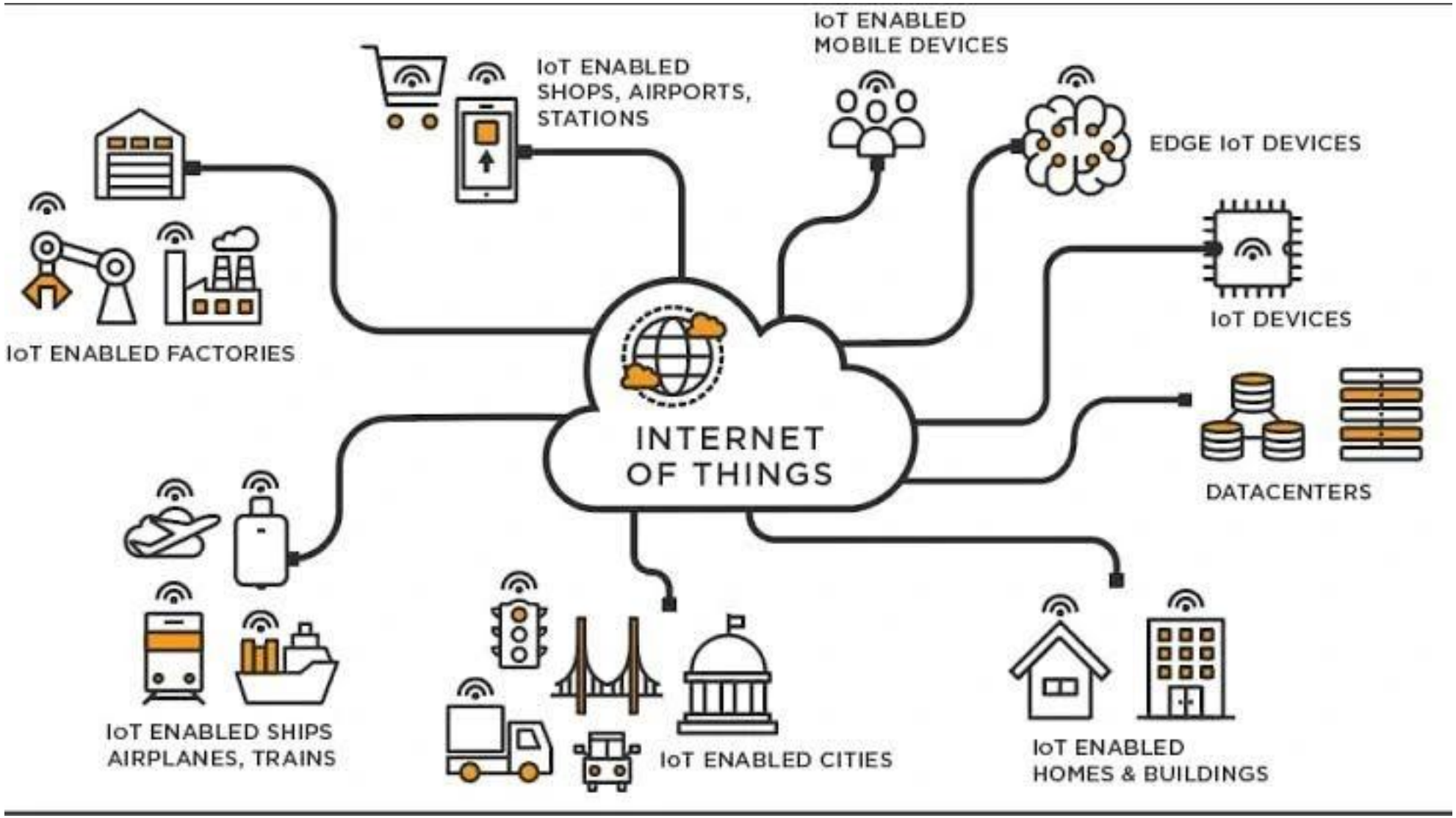
IOT Enabling Technologies

- IoT enabling technologies are the set of hardware, software, communication, and data-processing technologies that allow physical objects to sense, connect, exchange data, and act intelligently over the internet.



Source: https://www.outrightcrm.com/blog/internet-of-things-iot-enabling-technologies/
Source: https://www.includehelp.com/iot/iot-internet-of-things-enabling-technologies.aspx

IOT Enabling Technologies

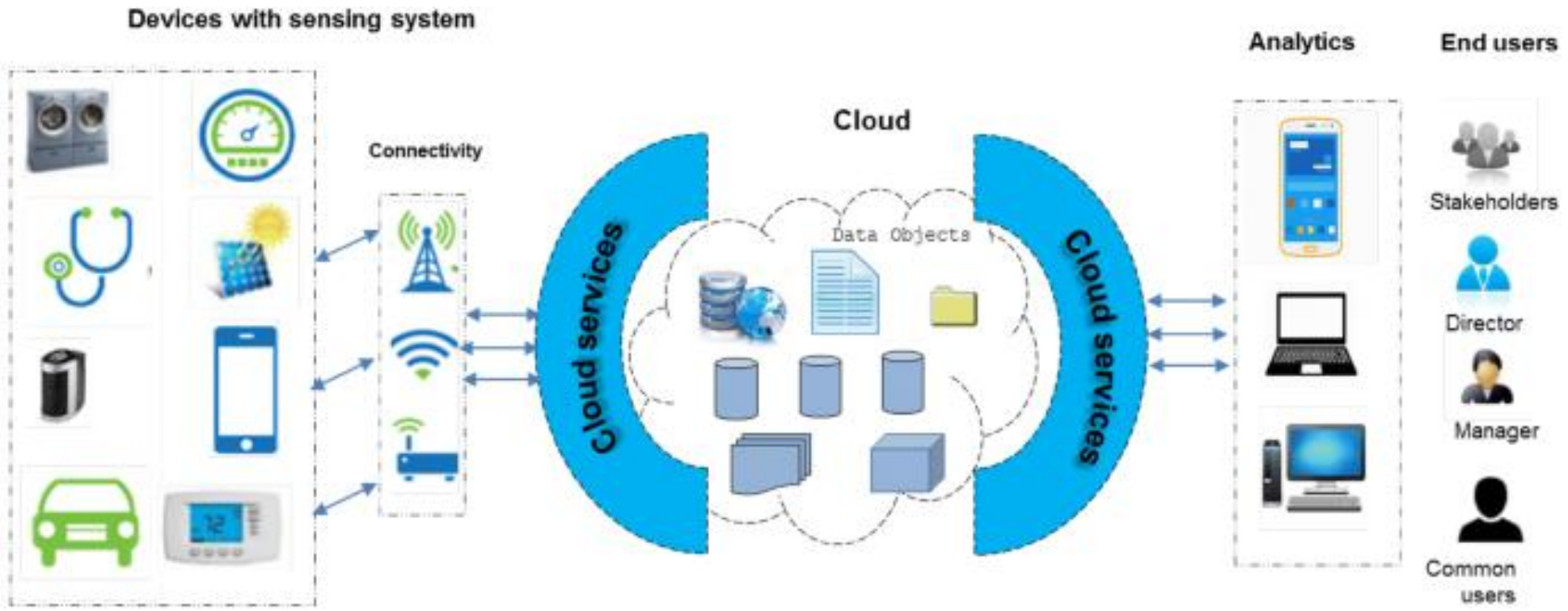


Source: https://runtimerec.com/embedded-industries/internet-of-things-iot/

IOT Enabling Technologies: Cloud Computing

- Cloud means something which is present in remote locations.

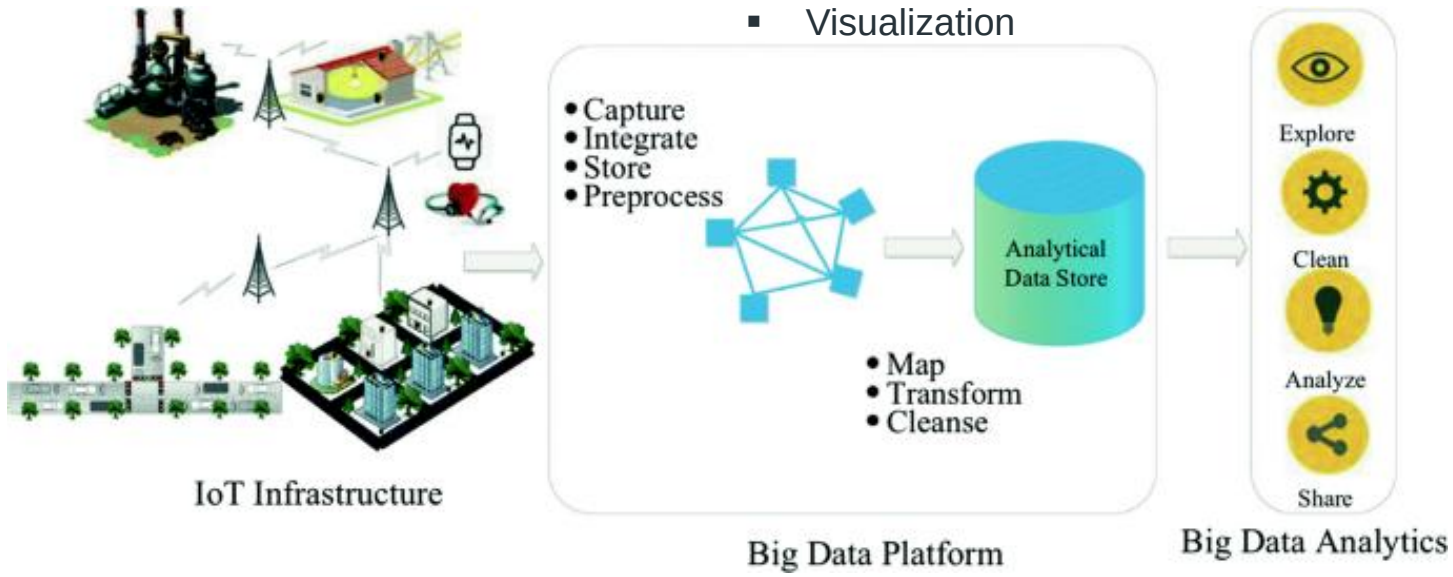
- With Cloud computing, users can access any resources from anywhere like databases, webservers, storage, any device, and any software over the internet.

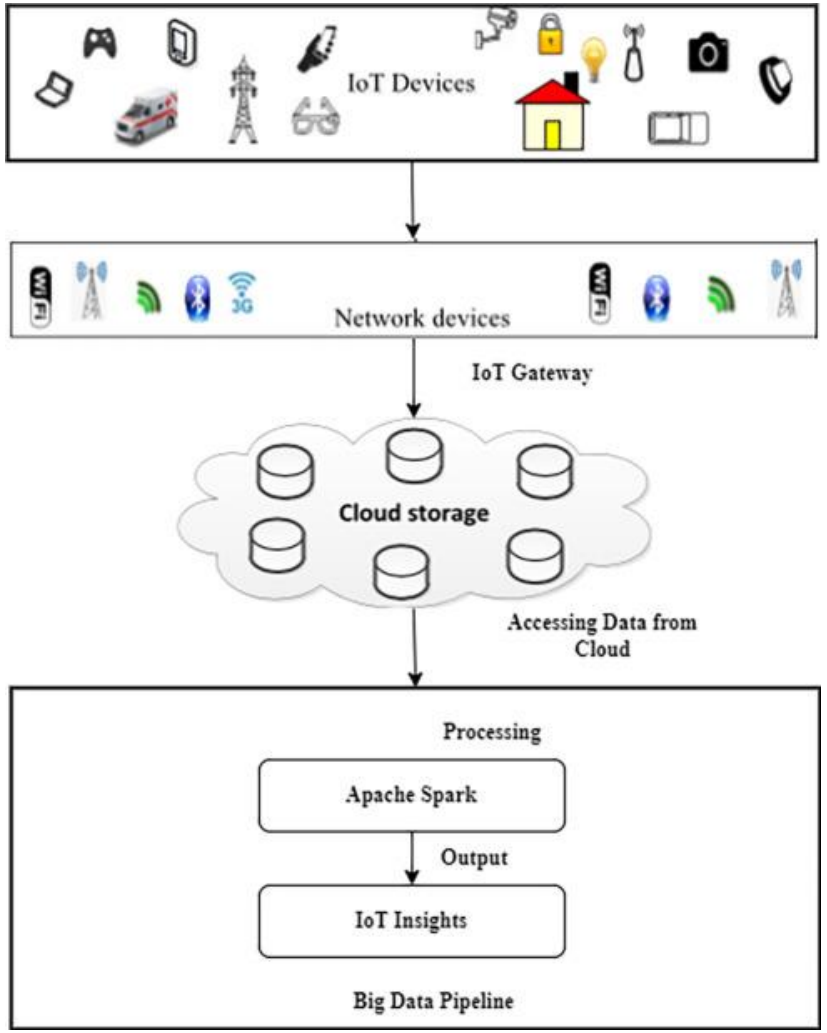


Source: https://doi.org/10.1007/s10586-018-2821-8

IOT Enabling Technologies: Big Data Analytics

- Big data is gathered from a variety of sources including social network videos, digital images, sensors and sales transaction records.
  - Data cleaning
  - Processing
  - Visualization





Examples

- Bank transactions
- Data generated by IoT systems for location and tracking of vehicles
- E-commerce
- Health and fitness data generated by IoT system using fitness bands

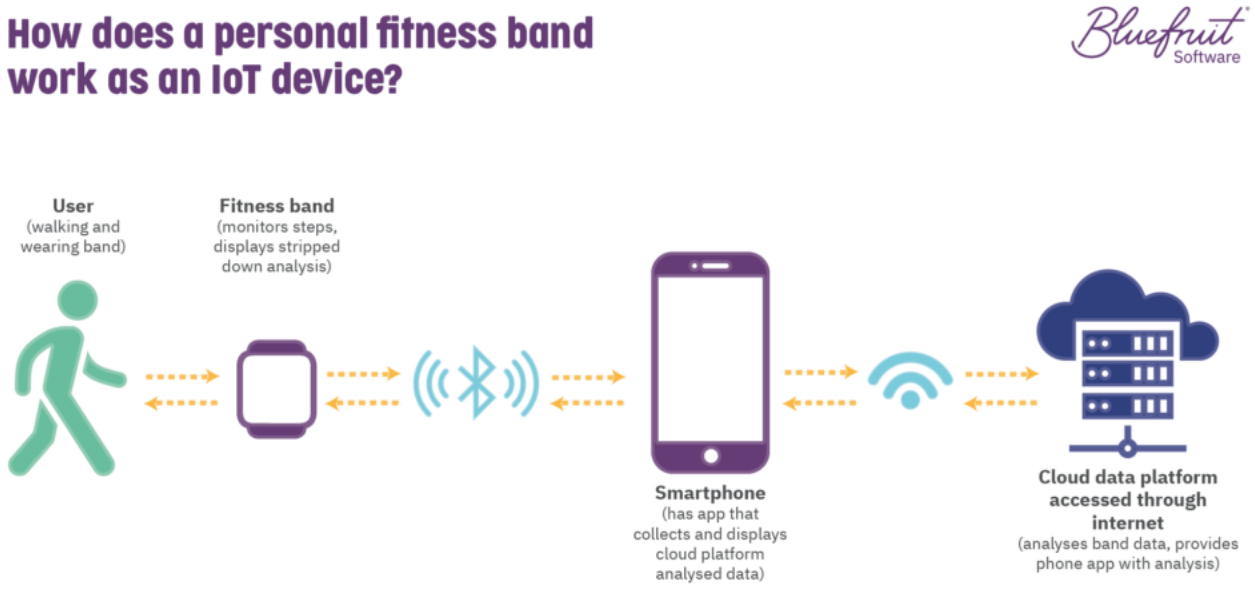
Source: https://link.springer.com/chapter/10.1007/978-3-030-41746-8_8
Source: https://www.geeksforgeeks.org/electronics-engineering/internet-of-things-iot-enabling-technologies/

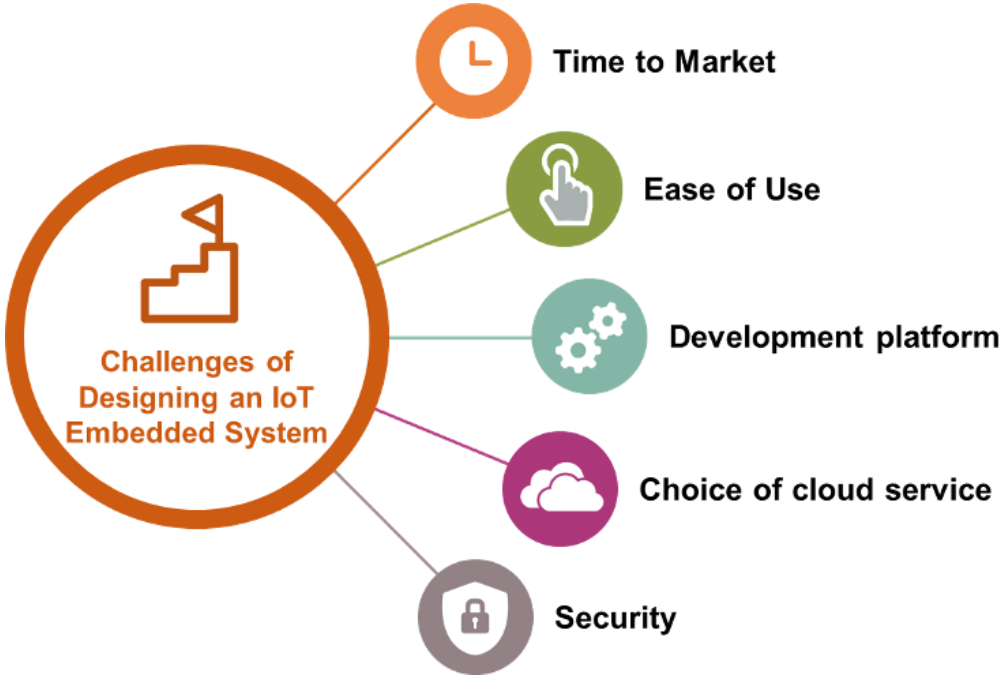
IOT Enabling Technologies: Embedded Systems

- Embedded systems are the backbone of IoT, enabling smart devices to communicate, process data, and interact with the cloud

Examples

- Digital camera
- DVD player, music player
- Industrial robots
- Wireless Routers





Source: https://bluefruit.co.uk/quality/your-guide-iot-and-embedded-systems/
Source: https://embeddedcomputing.com/technology/iot/edge-computing/simplifying-software-development-of-embedded-iot-applications

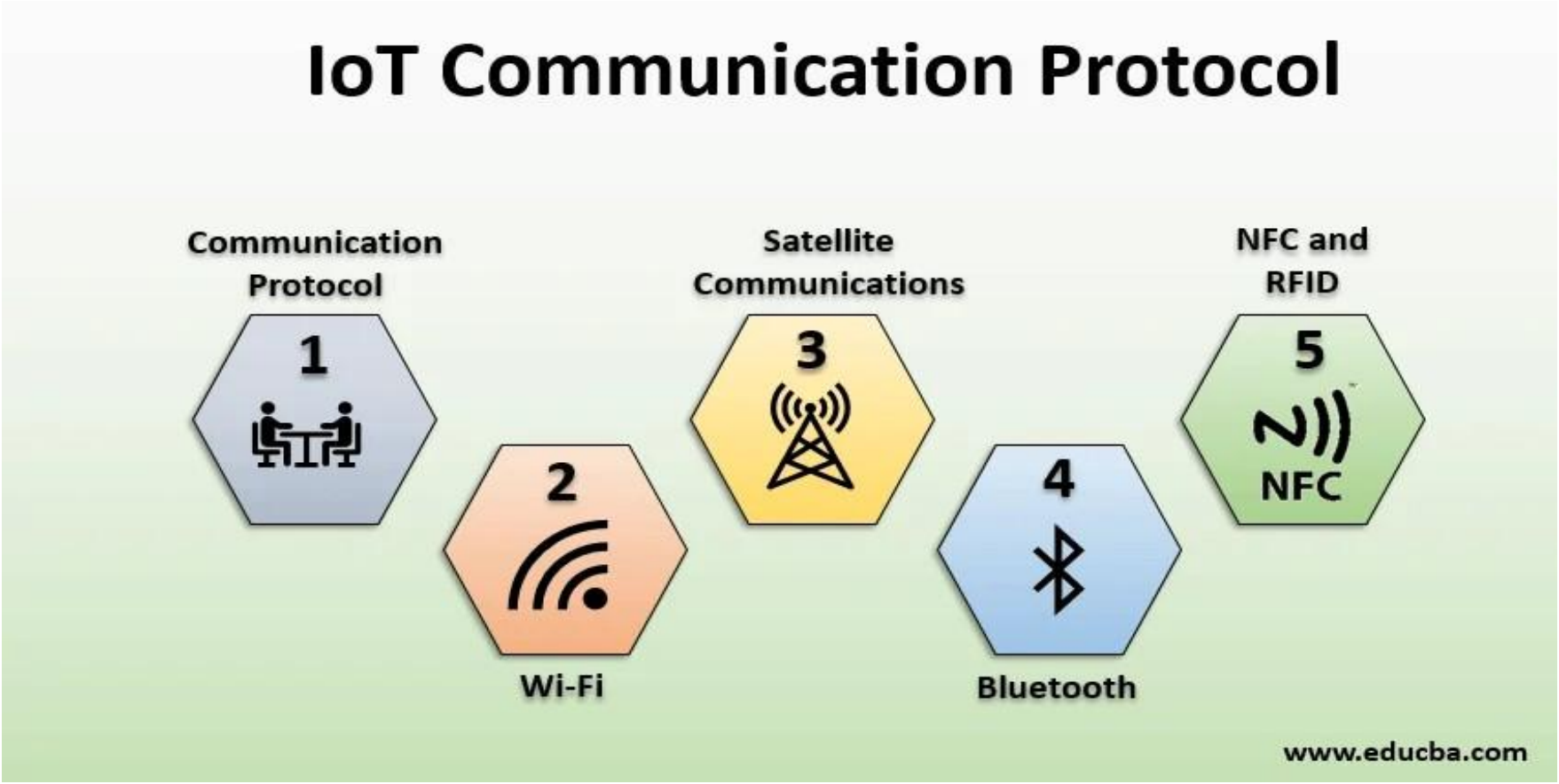
IOT Enabling Technologies: Communication

(How data is transmitted between IoT devices)

- Communication protocols allow devices to exchange data over the network

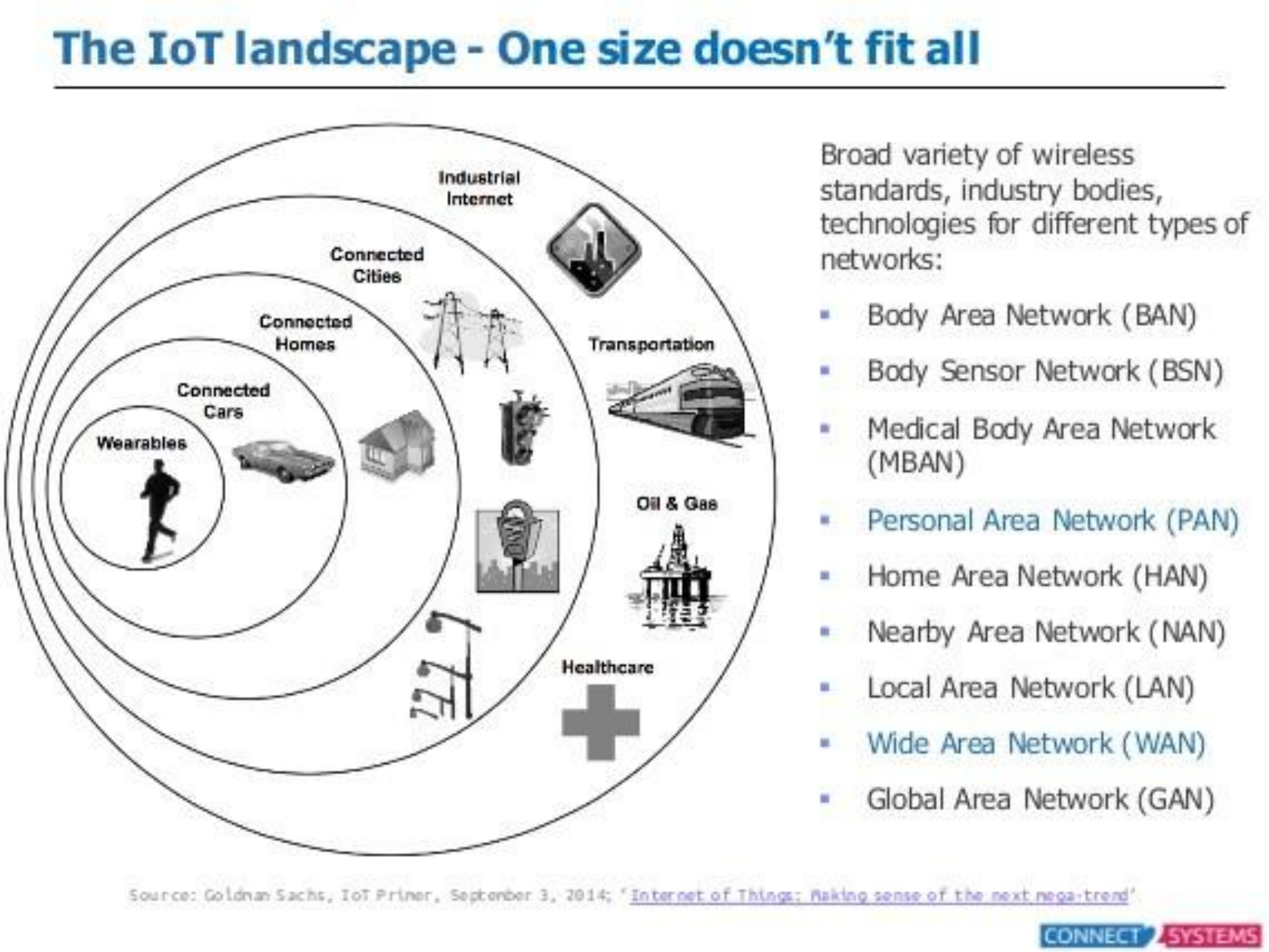
- How data is transmitted, received, and understood between two or more devices

- **Formatted**
- **Sent and received**
- How errors are **detected and corrected**
- How connection is **established and closed**



Source: https://www.educba.com/iot-communication-protocol/

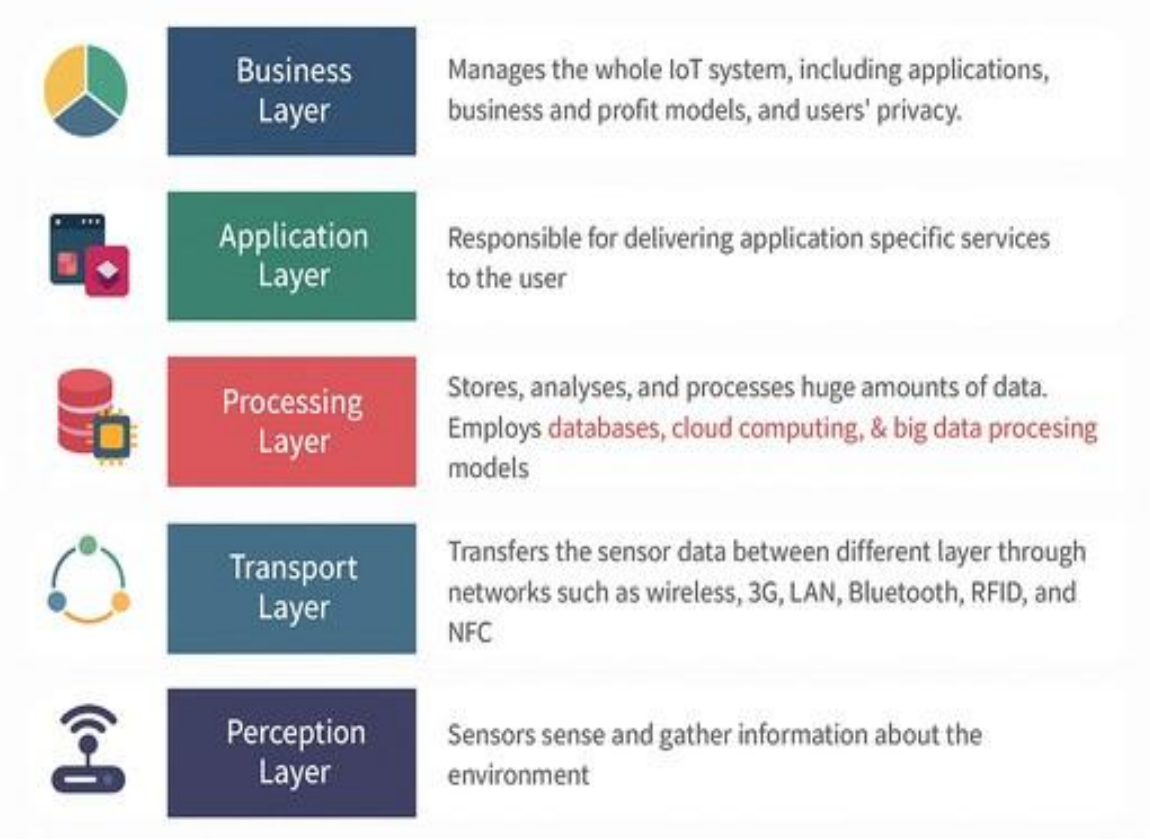
- IoT protocols are standards for communication and data transmission between connected IoT devices
- IoT standards are specific to IoT device requirements



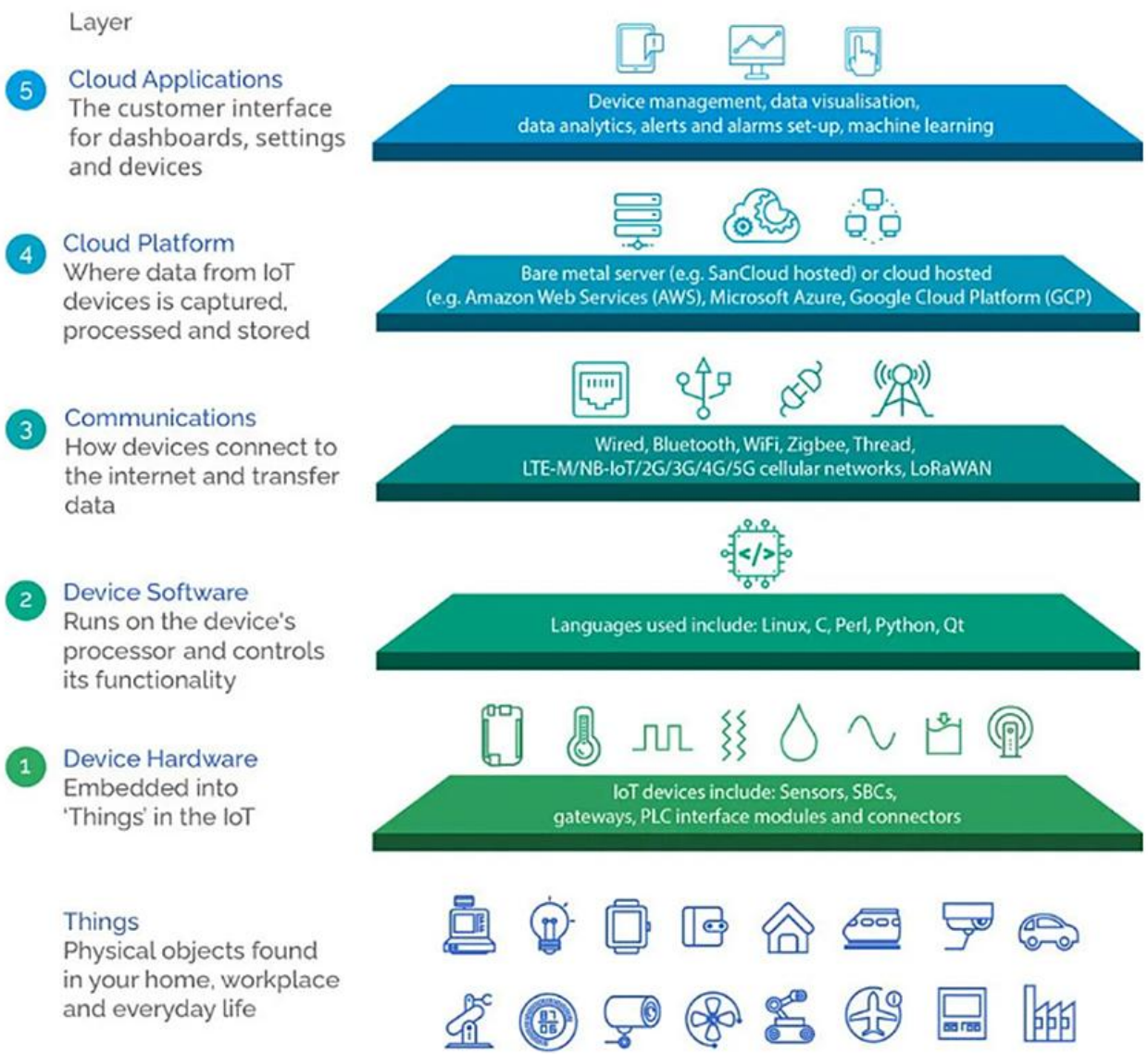
Source: https://www.gadgeon.com/blog/what-is-the-right-wireless-technology-for-your-iot-project/

IOT Technology Stack Diagram

- The IOT technology stack refers to the multiple layers of hardware, software and communication technologies that connect objects over the internet to monitor or control them

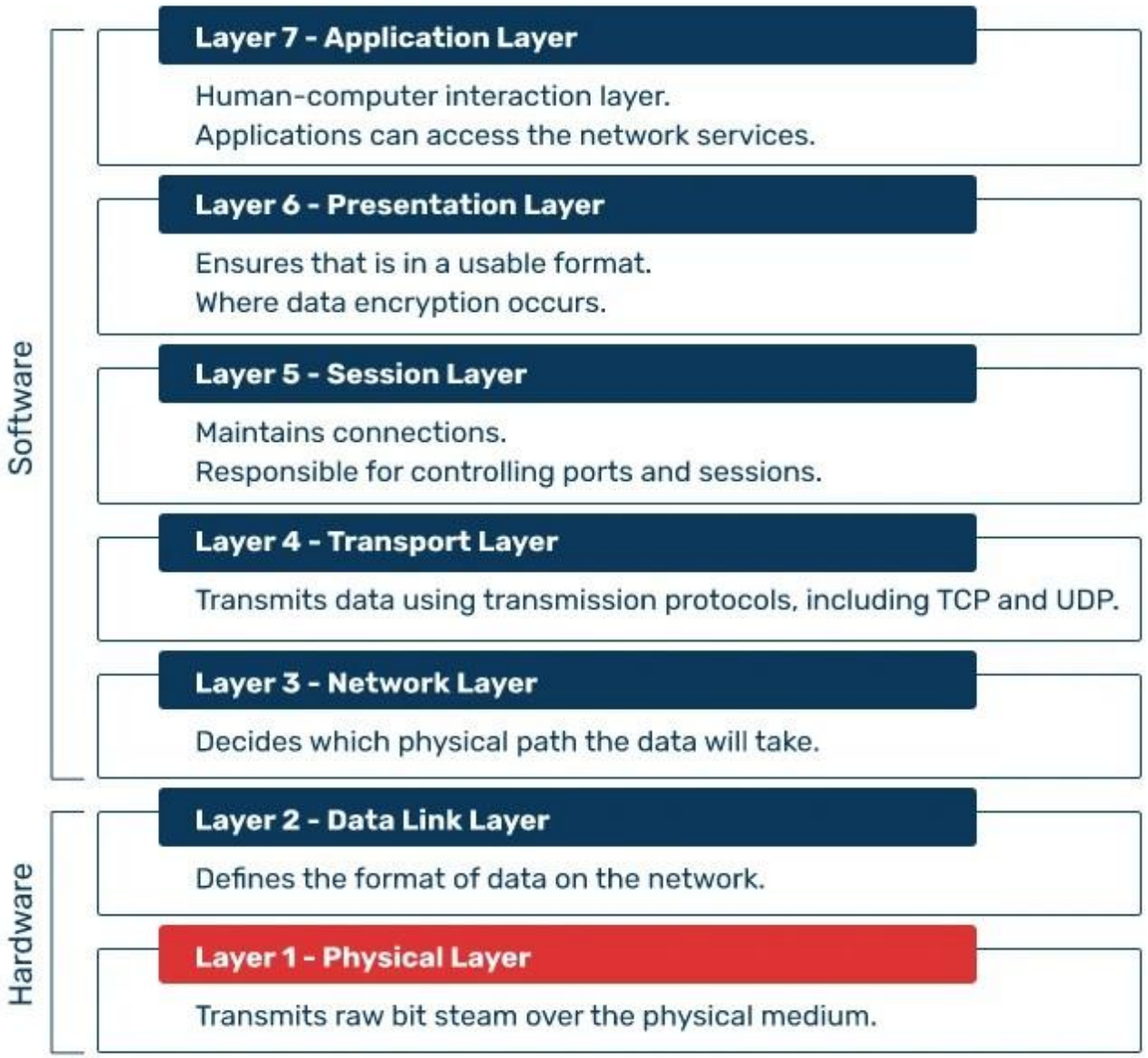


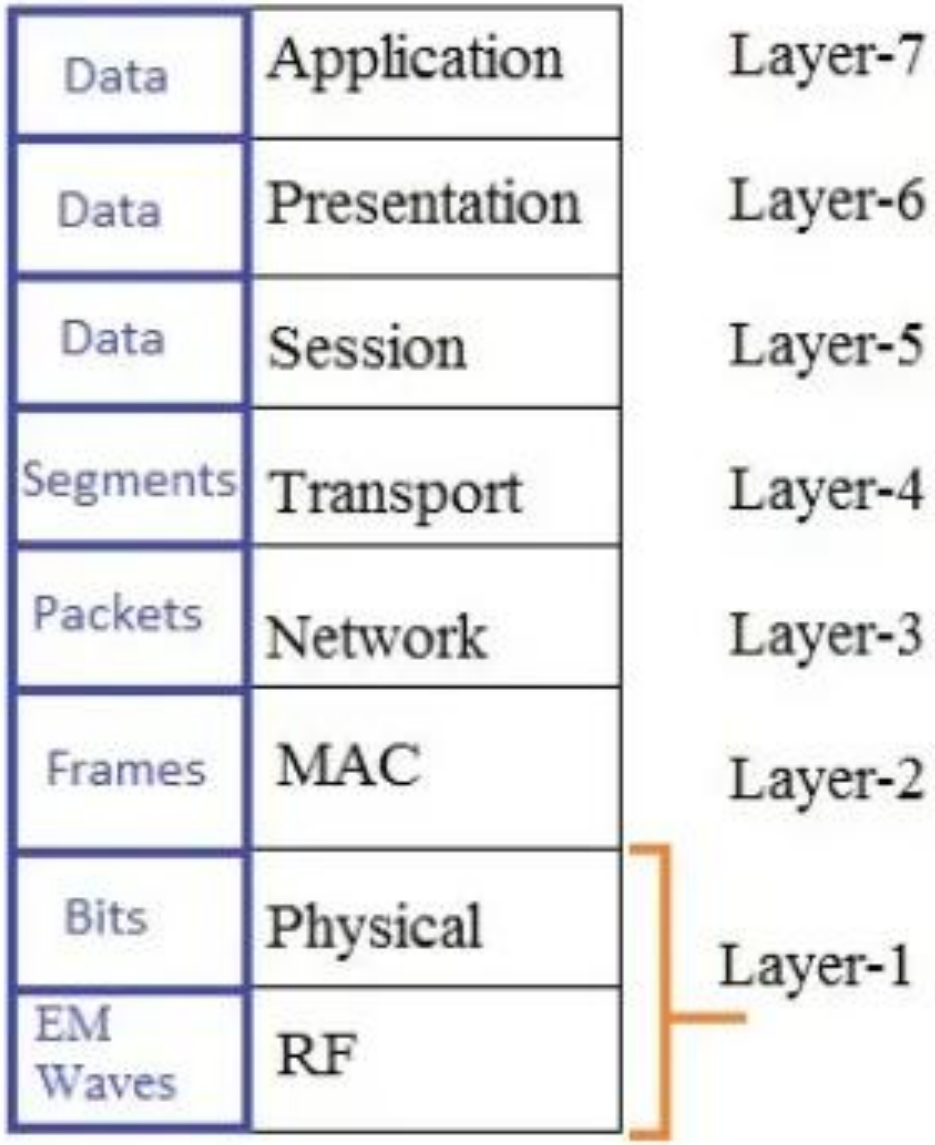
Source: https://www.eeworldonline.com/what-are-the-different-key-layers-of-iot-architecture-part-1/



Source: https://iotbusinessnews.com/2022/07/13/86750-what-is-the-iot-technology-stack/

IoT Layers Design



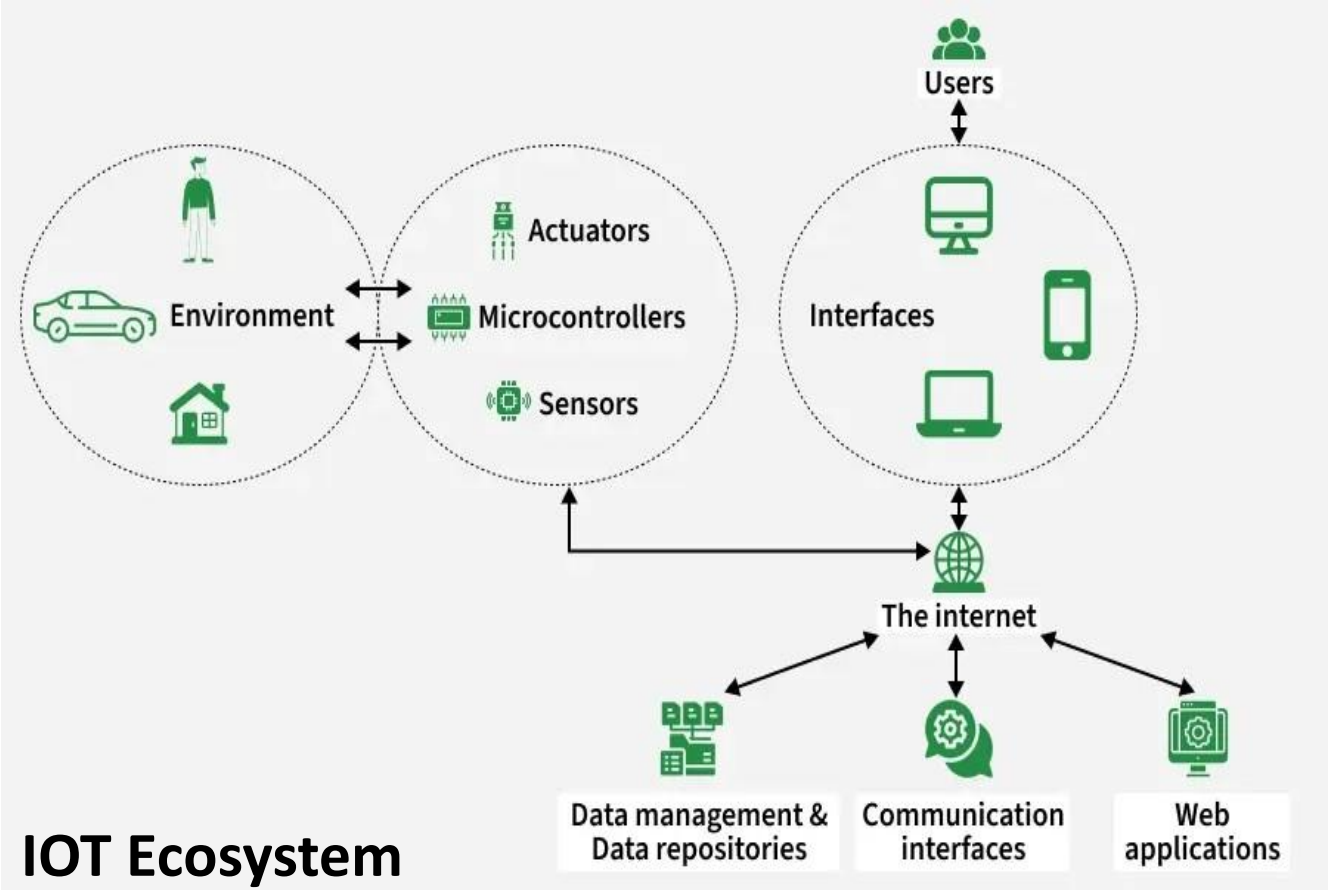


Source: https://sepiocyber.com/blog/osi-model/

Source: https://www.rfwireless-world.com/terminology/phy-vs-mac-in-iot-networking

Functional Block of an IOT Ecosystem

The Internet of Things (IoT) refers to a network of **interconnected physical objects** such as devices, machines, vehicles, or people embedded with sensors, software, and unique identifiers that enable them to collect, exchange, and process data over a network without requiring direct human-to-human or human-to-computer interaction.



Components of IoT System

1. Devices and Sensors

- Physical objects embedded with sensors or actuators that collect data from the environment (e.g., temperature, motion, gas, light).
- **Sensors** convert physical signals into digital data for further analysis.

2. Connectivity: Networks such as Wi-Fi, Bluetooth, or 5G that transmit data between devices, gateways, and cloud platforms.

3. Data Processing (transforms raw data into meaningful insights)

- Edge devices, gateways, or cloud platforms process and analyze collected data, often using AI or big data technologies

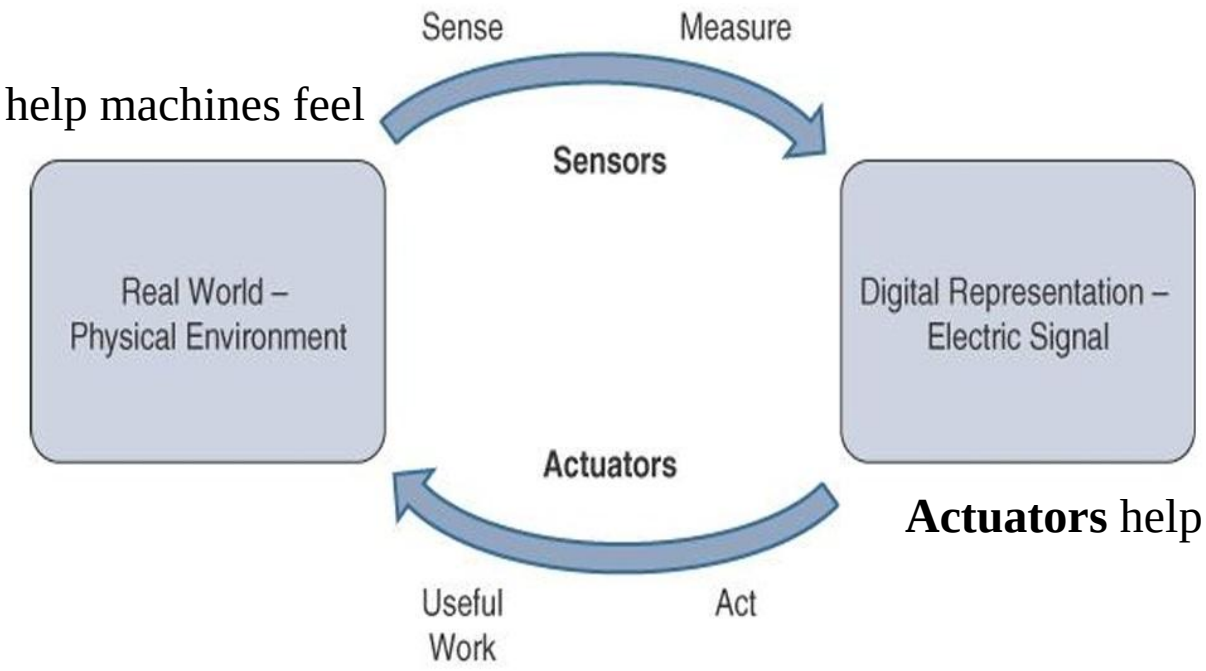
4. User Interface

- Applications, dashboards, or mobile apps through which users interact with IoT systems.
- Interfaces can also trigger actuators to perform automated actions.

Source: https://www.geeksforgeeks.org/computer-networks/introduction-to-internet-of-things-iot-set-1/

Sensor and Actuators

Sensors help machines feel



Actuators help them respond

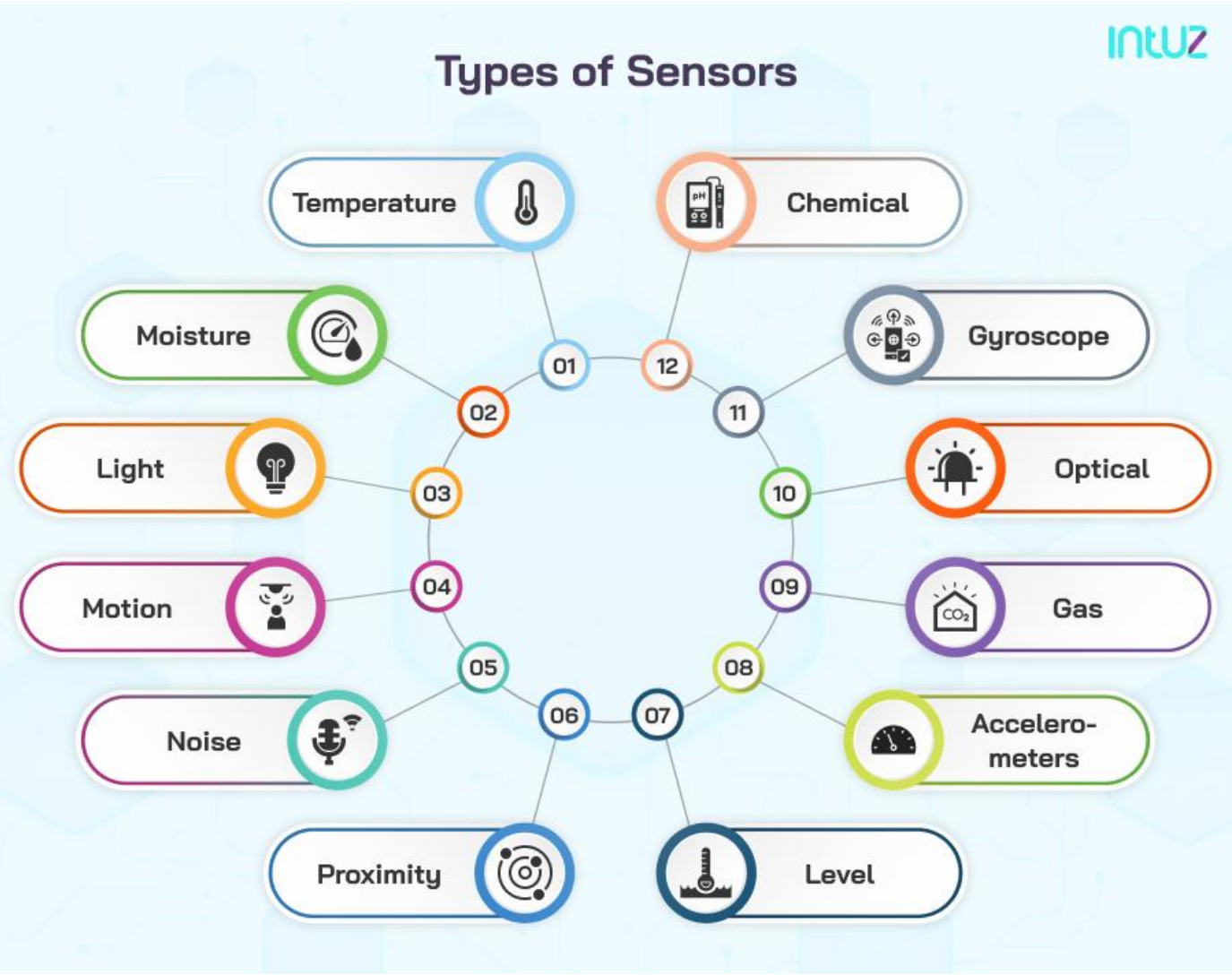
Source: https://www.atpinc.com/blog/What-is-iiot-vs-iot-actuators-edge-cloud-storage-sensor-data
Source: https://mite.ac.in/wp-content/uploads/2021/04/iot_module2.pdf

Sensors

- **Sensor** is a device that detects changes in physical parameters such as temperature, pressure, light, motion, gas, or humidity and converts them into measurable signals.

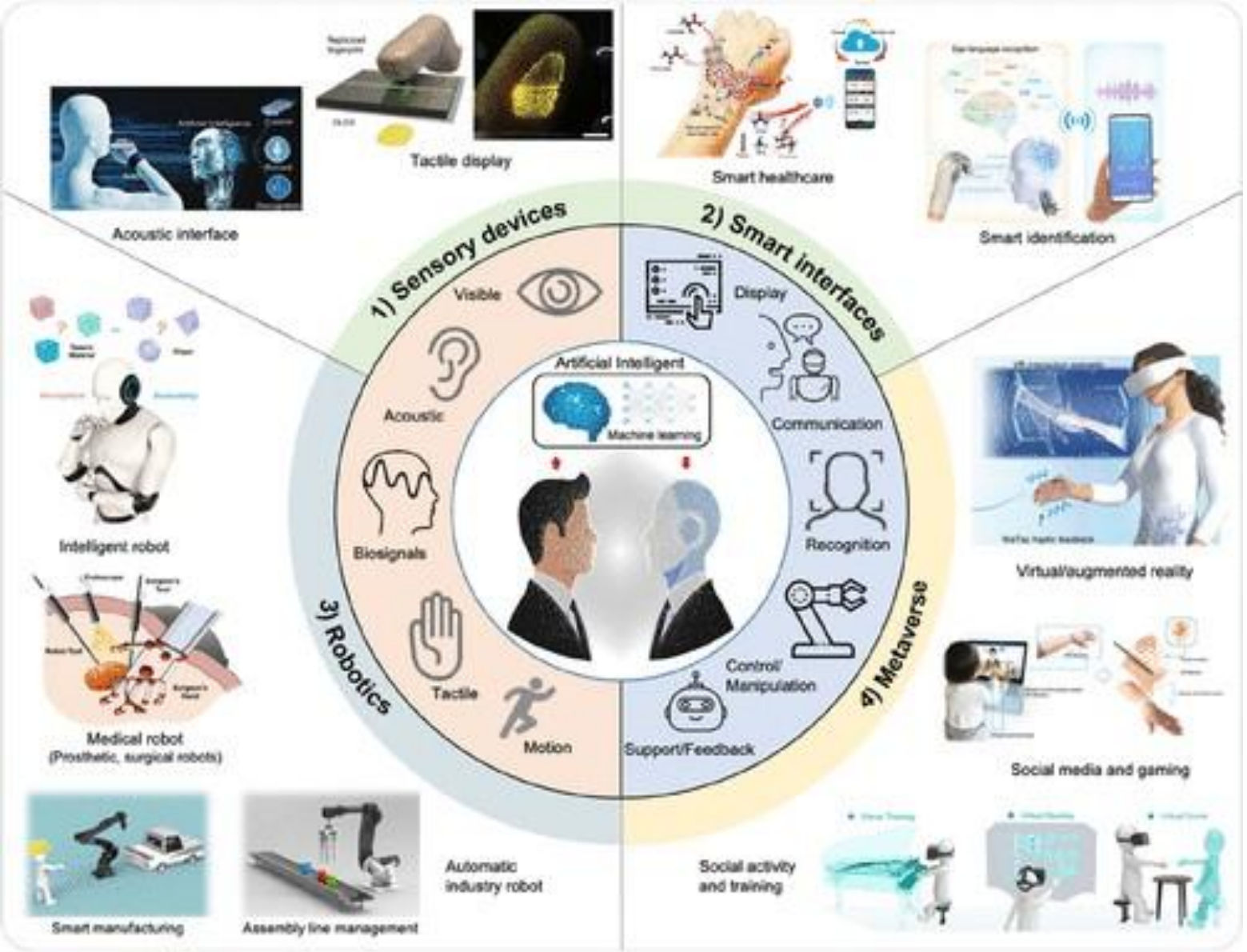
Why We Use Sensors

- To collect real-time data for monitoring and control
- To automate processes and reduce human effort
- To improve safety and prevent accidents
- To enhance efficiency and performance of systems
- To support decision-making using data
- To reduce energy consumption and operational cost
- To improve quality of life in healthcare, agriculture, and daily activities



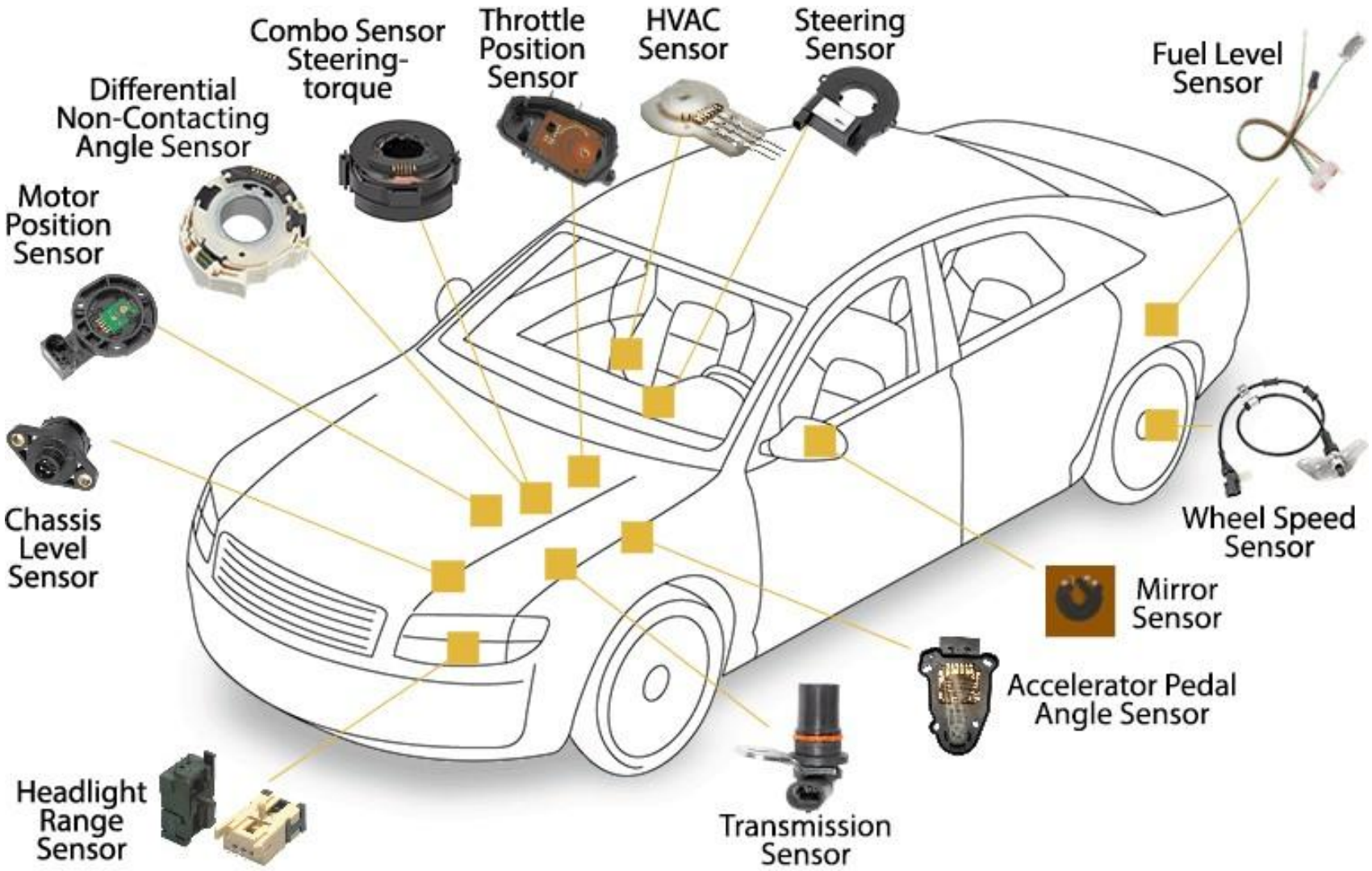
Source: https://dzone.com/articles/type-of-sensors-and-actuators-in-iot

Sensor Application in Real World



Source: https://doi.org/10.1021/acs.chemrev.3c00356

Sensor Application in Real World



Source: https://www.eedesignit.com/the-evolution-of-sensors-in-automotive-applications/

DIFFERENCE BETWEEN SENSOR AND ACTUATOR

SENSOR | **ACTUATOR**

Sensor vs Actuator

Difference between Sensor and actuator

SENSOR	ACTUATOR
Converts physical characteristics into electrical signals	Converts electrical signals into physical characteristics
Input from the environment	Output to environment
Output to input conditioning unit of the system	Input from the output conditioning unit of the systern
Generates electrical signals	Generates heat or motion Accepts commands to perform a function
Gives information to the system about the environment	Require an extra power source to operate
May or may not require an extra power source e to operate	Require an extra power source to operate Example: Electric motor

Real-World Examples

Smart Home

Sensor
Motion sensor detects movement.

Actuator
Smart light turns on automatically.

Healthcare

Sensor
Heart rate monitor tracks beats.

Actuator
Pump adjusts medication flow

Automotive

Sensor
Ultrasonic sensor for parking.

Actuator
Irrigation pump switches on when soil i dry.

Agriculture

How Sensors and Actuators Work Together in IoT

Sensor Reads Data
↓
Microcontroller Processes Data
↓
Decision Taken Based on Logic
↓

Example: Smart Fan System

Sensor → Reads room temperature

Microcontroller → Compares with threshold

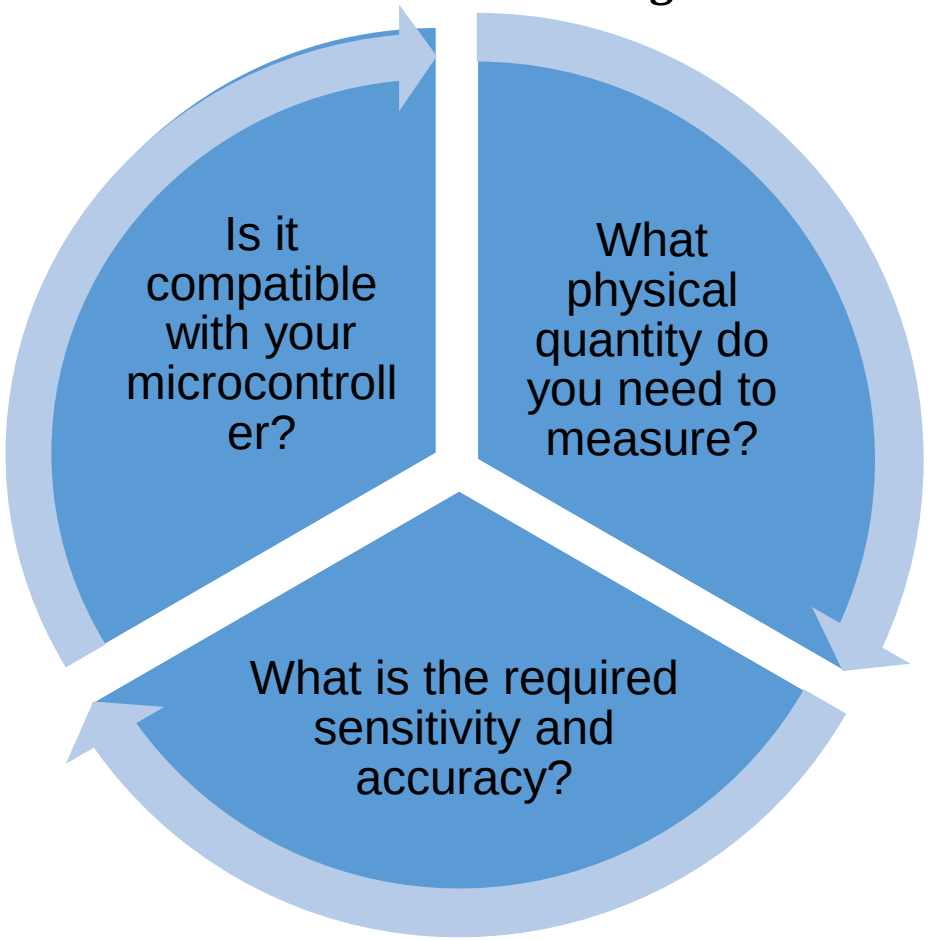
Actuator → Turns fan ON if temperature is high

Sensor → Reads room temperature
Actuator → Tompares weith ttthreshold

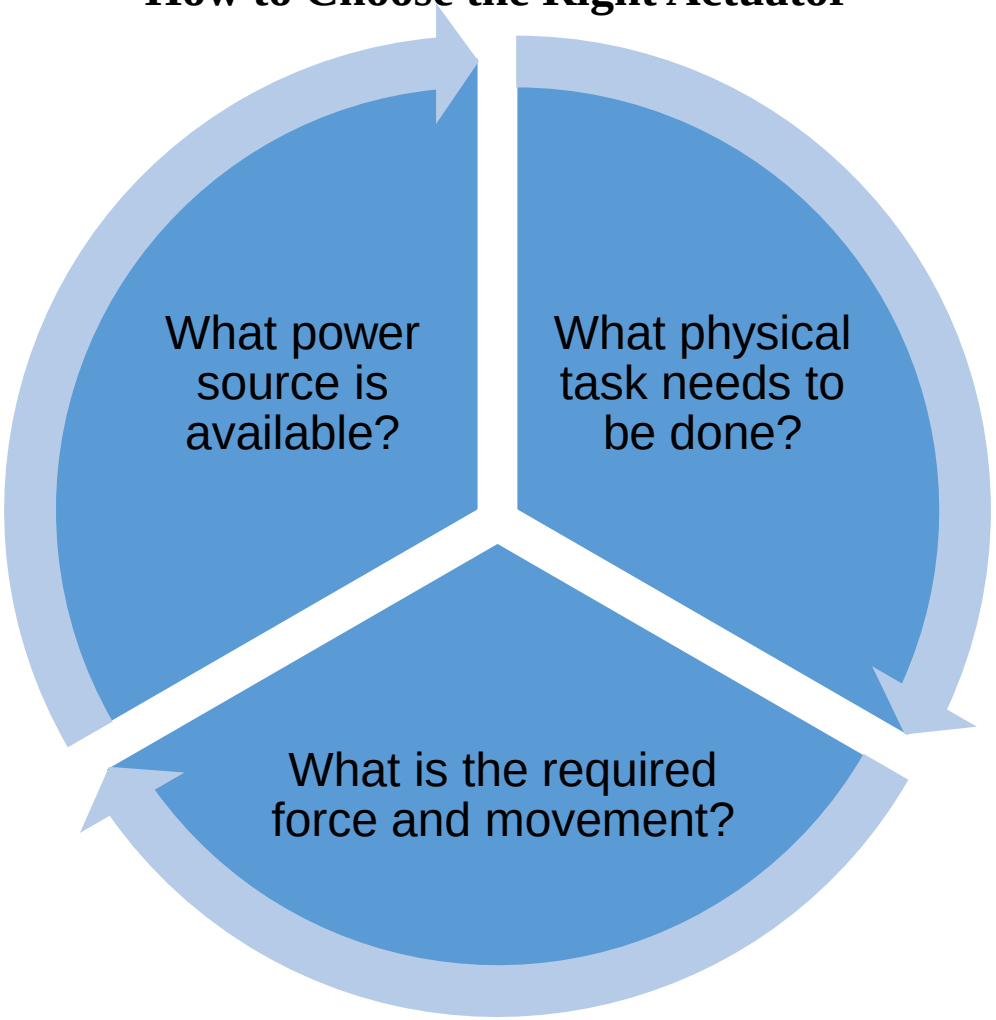
Source: https://iotdunia.com/difference-between-sensor-and-actuator/rbsnp/

Sensor and Actuators Selection Guide

How to Choose the Right Sensor



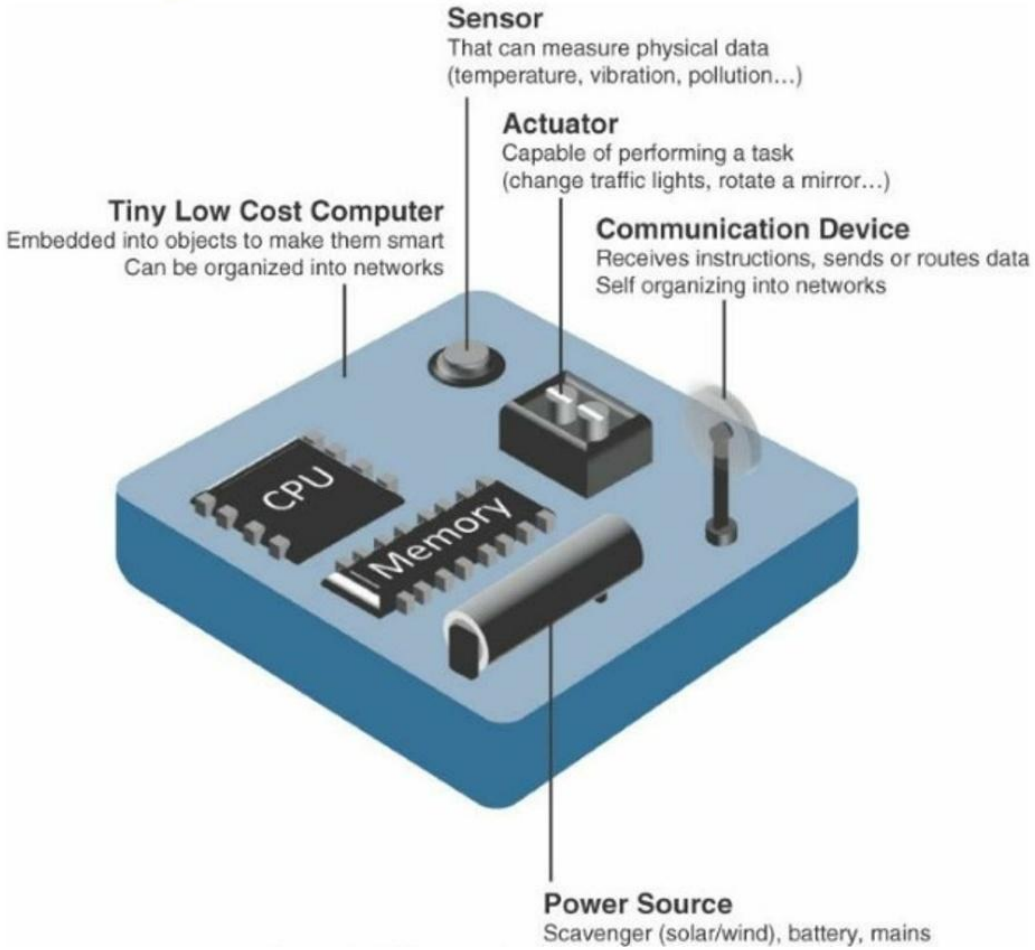
How to Choose the Right Actuator



Source: https://iotdunia.com/difference-between-sensor-and-actuator/rbsnp/

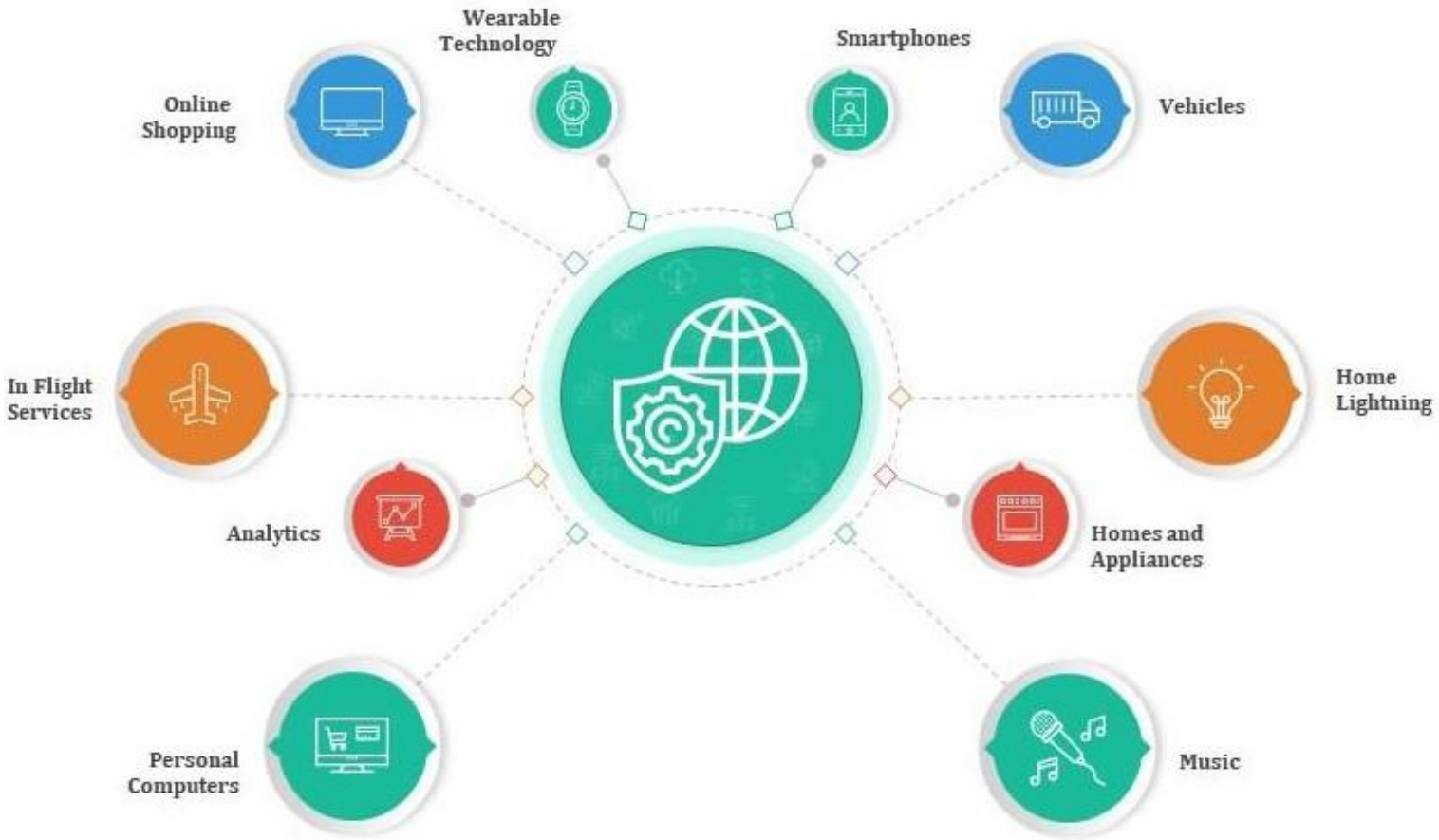
Smart Object

- A smart object in IoT is a physical object with **sensing**, **processing**, and **communication** capabilities that allow it to interact intelligently within an IoT system.



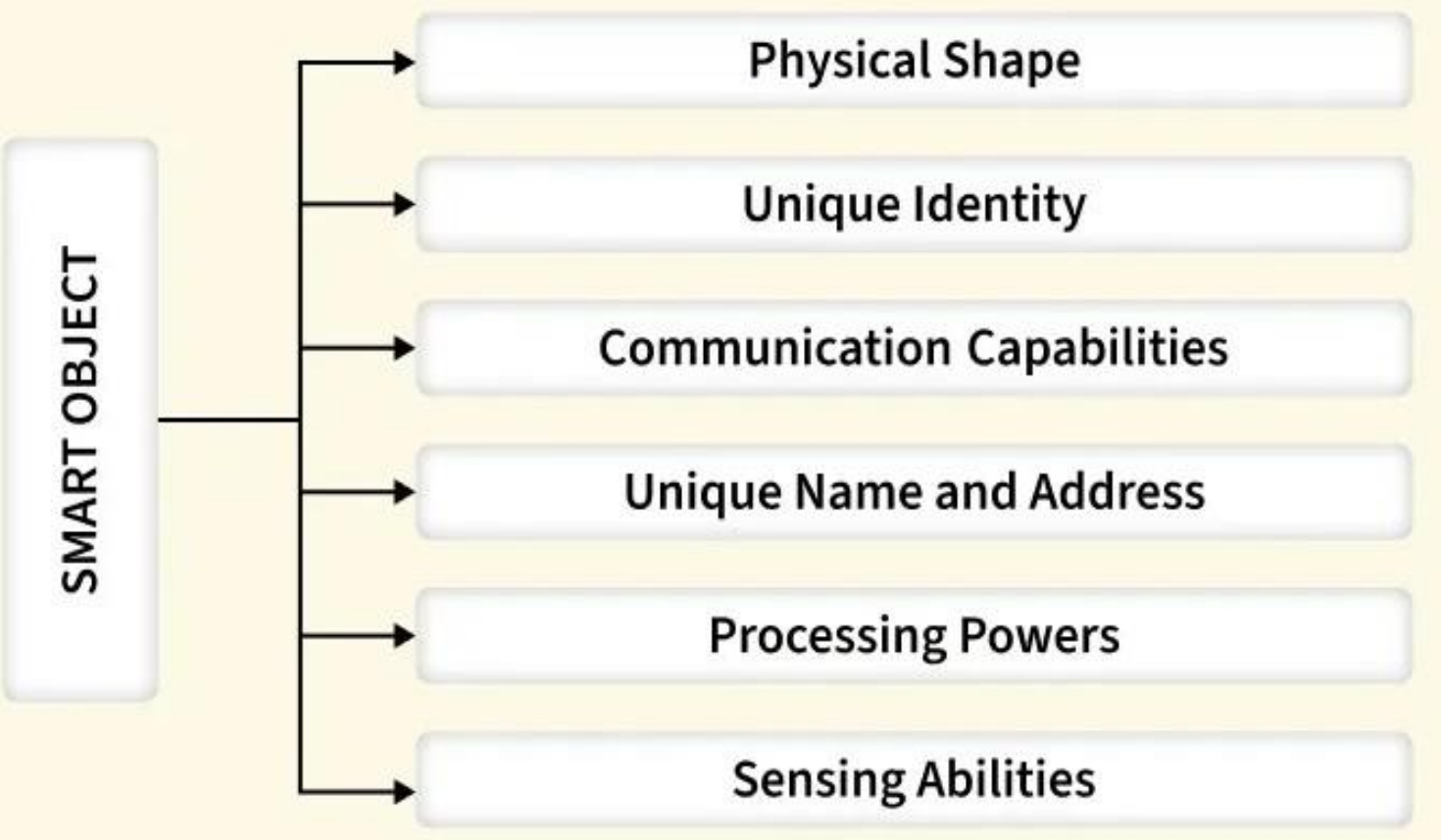
Source: https://www.slideteam.net/what-are-smarts-objects-in-iot-internet-of-things-iot-overview-ppt-powerpoint-presentation-icon.html

What are Smarts Objects in IOT ?



Source: https://www.geeksforgeeks.org/computer-networks/understanding-things-in-internet-of-things/

Characteristics of Smart Object



Physical shape refers to the size, form, and structure of a smart object, which affects how it is used

Each smart object has a unique ID (IP address) for identification and tracking

Can communicate with other devices or systems using wired or wireless technologies (Wi-Fi, Bluetooth)

Contains embedded processing units (microcontroller/microprocessor) to analyze data locally

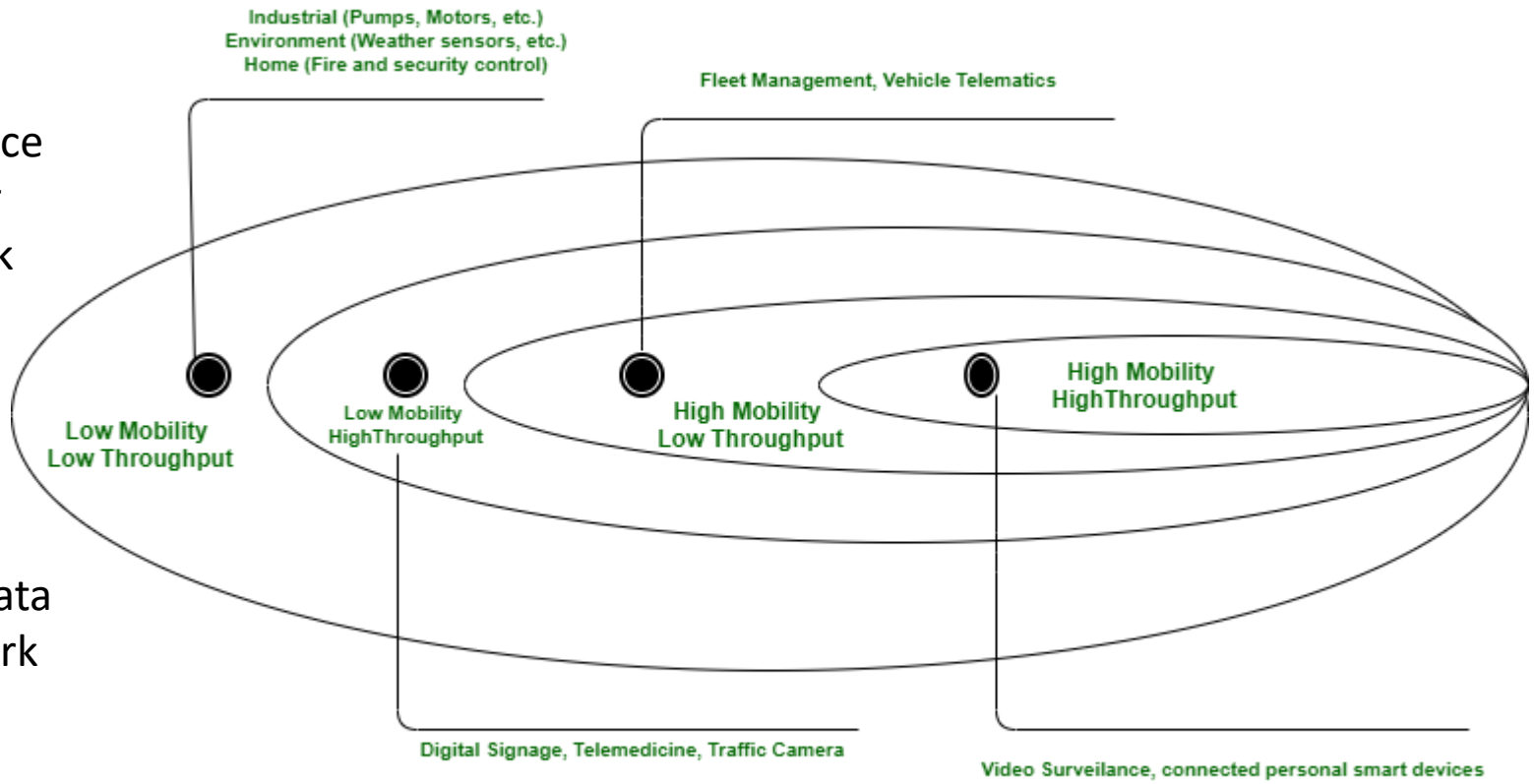
Equipped with sensors to collect data from the physical environment

Source: https://www.slideserve.com/mimis/domestic-logistics-management-services-dlms

Smart Objects

- Mobility is the ability of a user or device to move from one location to another while maintaining continuous network connectivity

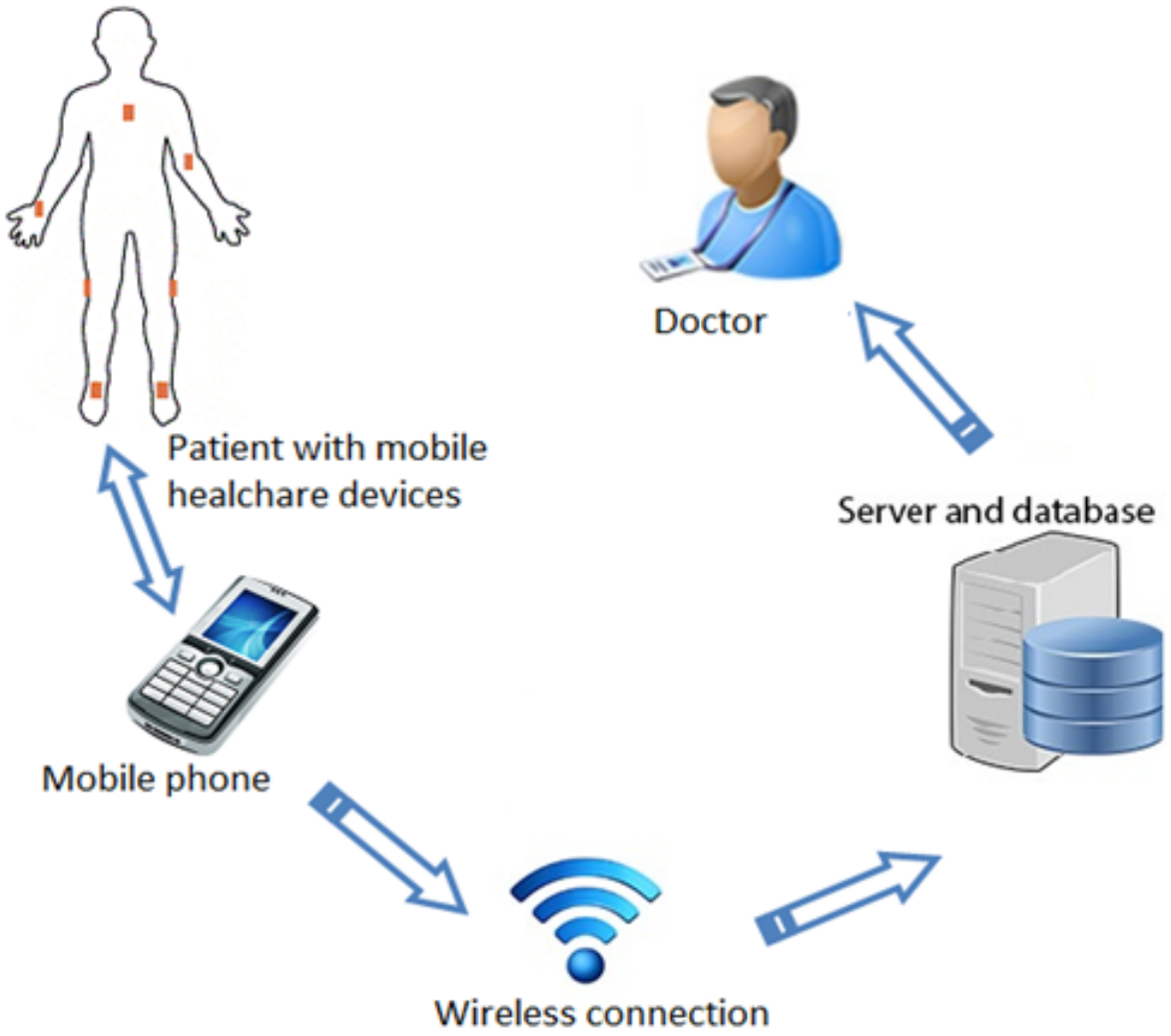
- Throughput is the actual amount of data successfully transmitted over a network per unit time



An Example of Sensor applications based on mobility and throughput

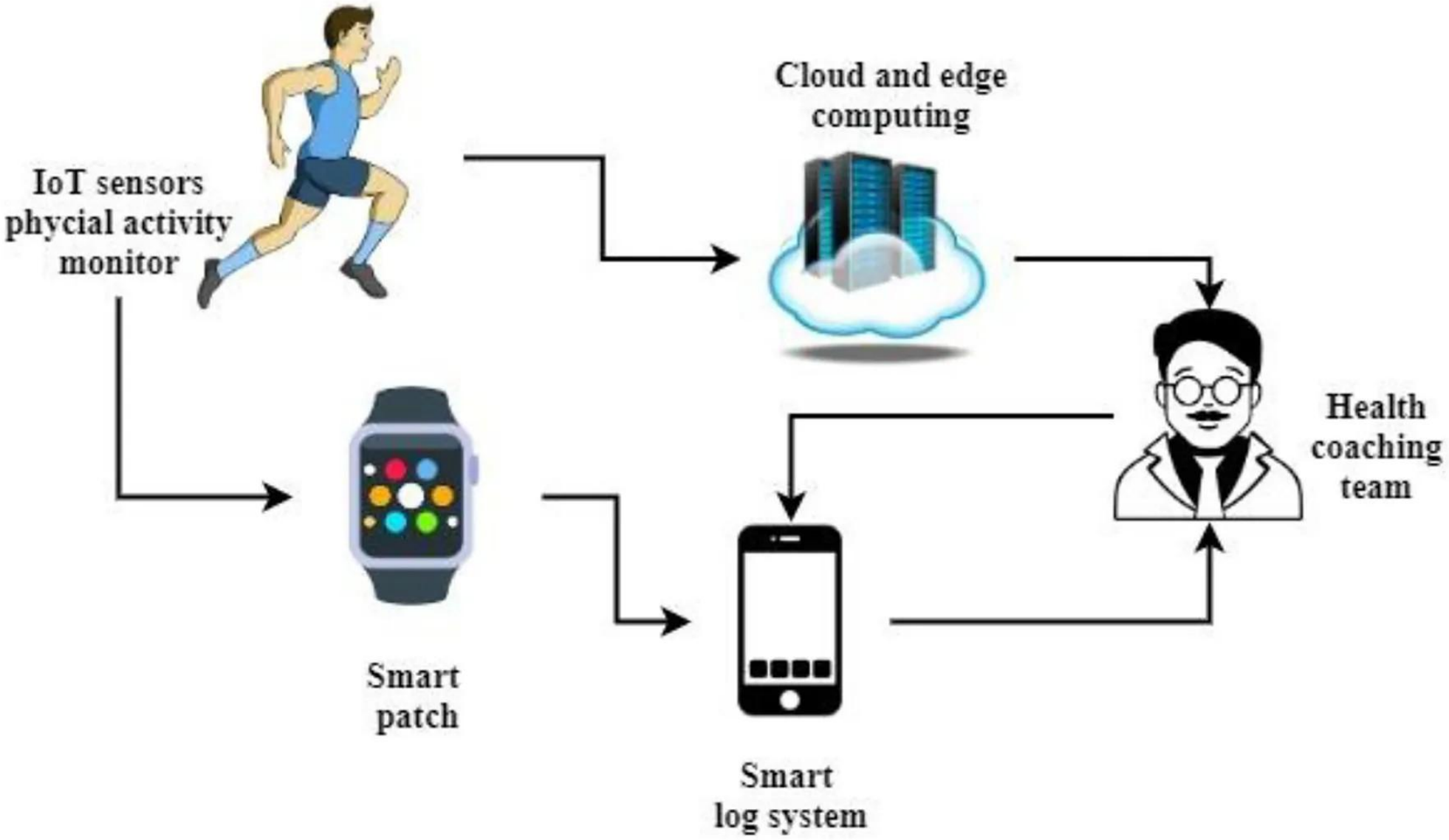
Source: https://www.geeksforgeeks.org/computer-organization-architecture/classification-of-smart-objects/

Connecting Smart Objects in Healthcare



Source: 10.5220/0004913905110516

Connecting Smart System in Healthcare



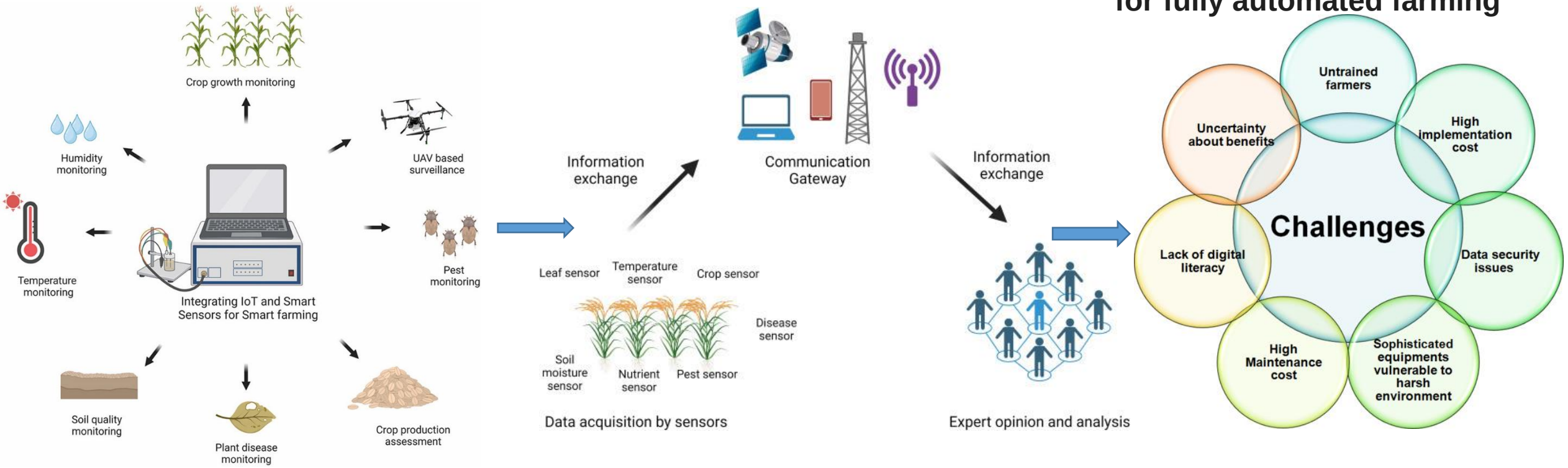
Source: https://www.researchgate.net/figure/Using-IoT-architecture-in-health-monitoring-systems_fig2_371248558

Connecting Smart Objects in Agriculture

Applications of integrated IoT and smart sensors for precision farming

Integration of IoT devices farms, and human networks

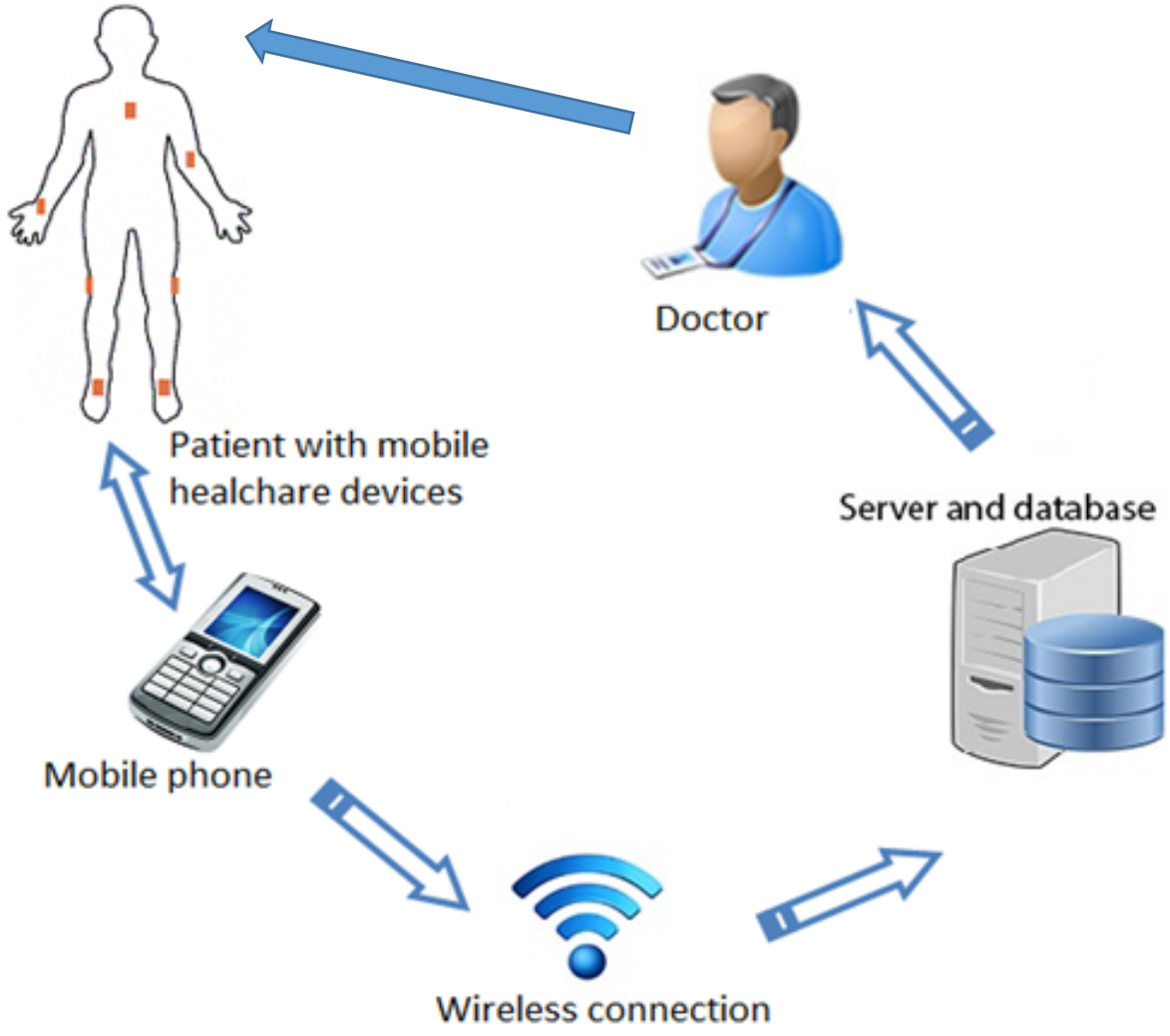
Different challenges encountered with regard to the deployment of IoT-based smart sensors and accessory devices for fully automated farming



Source: https://doi.org/10.1016/j.jafr.2023.100776

Smart Device

Connecting Smart System in Healthcare



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Thank
you